PHYSICS

1) A small object is dropped from the top of a building and falls to the ground. As it falls, accelerating due to gravity, it passes a window. It has speed v_1 at the top of the window and speed v_2 at the bottom of the window, at what point does it have a speed $\frac{v_1 + v_2}{2}$? Neglect the air resistance.

(1) It depends on the height of the window or its distance from the top of the building.

(2) Above the centre point of the window.

(3) Below the centre point of the window.

(4) At the centre point of the window.

2) An object is moving in x–y plane with the position as a function of time is given by $\vec{r} = x(t)\hat{i} + y(t)\hat{j}$. Point 'O' is at origin having x = 0 and y = 0. The object is definitely moving towards origin 'O' when :-

(1) $\frac{dx}{dt} > 0, \frac{dy}{dt} > 0$

(2) $\frac{dx}{dt} < 0, \frac{dy}{dt} < 0$

(3) $x\frac{dx}{dt} < 0, y\frac{dy}{dt} < 0$

(4) $x\frac{dx}{dt} > 0, y\frac{dy}{dt} > 0$

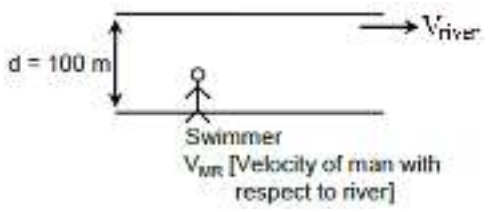
3) A particle is projected from ground with an initial speed $4\sqrt{7}$ m/s at an angle 60º with horizontal. The average velocity of the particle between its point of projection and highest point of trajectory is.

(1) 5 m/s

(2) 6 m/s

(3) 7 m/s

(4) 8 m/s



4) Choose correct option(s) : (A) if V_{river} = 8 m/sec & V_{MR} = 10 m/sec then time taken by swimmer to cross the river along shortest path is $\frac{50}{3}$ sec.

(B) if V_{river} = 8 m/sec & V_{MR} = 10 m/sec then minimum time taken by swimmer to cross the river is 10 sec.

(C) if V_{river} = 10 m/sec & V_{MR} = 8 m/sec then zero drift cannot be possible

(D) if V_{river} = 10 m/sec & V_{MR} = 8 m/sec then for minimum drift swimmer tries to swim at an angle of

143° with the downstream.

(1) Only B

(2) A,B and C

(3) All correct

(4) All are incorrect

5) A block A of mass 4 kg is placed on another block B of mass 5 kg, and the block B rests on a smooth horizontal table. If the maximum force that can be applied on A so that both the blocks move together is 12N, the maximum force that can be applied to B for the blocks to move together will be ..... N :

(1) 30

(2) 25

(3) 15

(4) 48

6) Assuming the coefficient of friction between the road and tyres of a car to be 0.5, the maximum speed with which the car can move round a curve of 40.0 m radius without slipping, if the road is unbanked, should be ..... m/s :-

(1) 25

(2) 19

(3) 14

(4) 11

7) Inertia is that property of a body by virtue of which the body is :-

(1) Unable to change the state of rest by itself

(2) Unable to change the state of uniform motion by itself

(3) Unable to change the direction of motion by itself

(4) Unable to change the state of rest and of uniform linear motion by itself

8) A body of mass m is moving in a circle of radius r with a constant speed v. The force on the body is $\frac{mv^2}{r}$ and is directed towards the centre. What is the work done by this force in moving the body over half the circumference of the circle -

(1) $\frac{mv^2}{\pi r^2}$

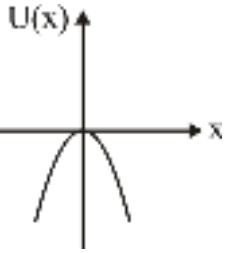
(2) Zero

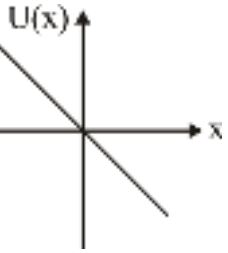
(3) $\frac{mv^2}{r^2}$

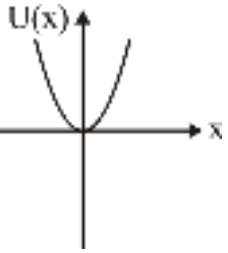
(4) $\frac{\pi r^2}{mv^2}$

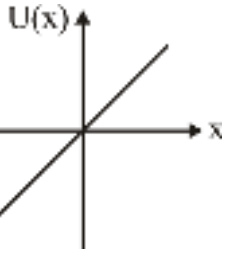
9) A particle is placed at the origin and a force F = kx is acting on it (where k is positive constant). If

U(0) = 0, the graph of U(x) versus x will be (where U is the potential energy function)

(1) 

(2) 

(3) 

(4) 

10) In a new system of units force (F), momentum (P) and mass (M) are taken as fundamental physical quantities. Find the dimensional formula of power in new system.

(1) FP^2M^{-1}

(2) F^0PM^{-1}

(3) FPM^{-1}

(4) $F^0P^2M^{-1}$

11) ABCD is a quadrilateral. Forces $\vec{BA}$, $\vec{BC}$, $\vec{CD}$ & $\vec{DA}$ act at a point. Their resultant is :

(1) $2\,\vec{AB}$

(2) $2\,\vec{DA}$

(3) zero vector

(4) $2\,\vec{BA}$

12) A body of mass m = 3.513 kg is moving along the x-axis with a speed of 5.00 m/s. The magnitude of its momentum is :

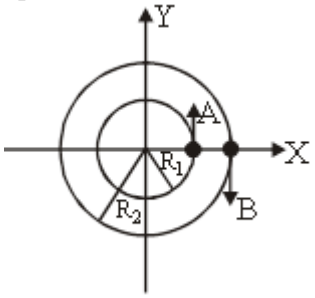
(1) 17.6 kg ms^{-1}

(2) 17.56 kg ms^{-1}

(3) 17.565 kg ms^{-1}

(4) 17.57 kg ms^{-1}

13) Two particles A, B are moving on two concentric circles of radii R_1 and R_2 with equal angular speed ω. At t = 0, their positions and direction of motion are shown in the figure :



The relative velocity $\vec{v}_A - \vec{v}_B$ at $t = \frac{\pi}{2\omega}$ is given by :

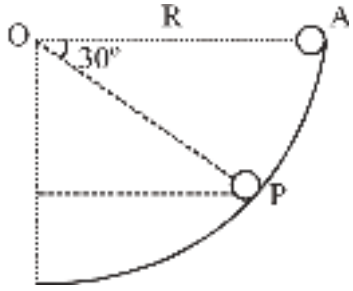
(1) $-\omega(R_1 + R_2)\hat{i}$

(2) $\omega(R_1 + R_2)\hat{i}$

(3) $\omega(R_1 - R_2)\hat{i}$

(4) $\omega(R_2 - R_1)\hat{i}$

14) A particle of mass m slides on a quarter part of smooth sphere of radius R as shown in figure. It is released from rest at A, the normal contact force exerted by surface on particle when it reaches P is:-



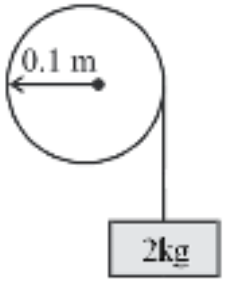
(1) $\frac{mg}{2}$

(2) $\frac{3mg}{2}$

(3) $mg \times \frac{\sqrt{3}}{2} + mg$

(4) $\frac{mg\sqrt{3}}{2}$

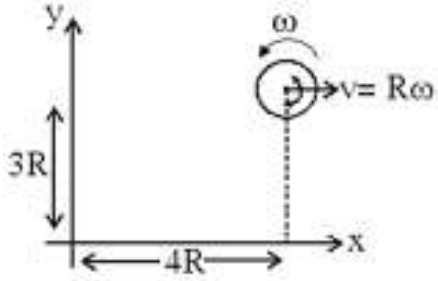
15) The moment of inertia of a solid flywheel about its axis is 0.1 kg-m^2. A block of 2kg is hanged round the circumference of the flywheel with the help of a string as shown in the figure. If the radius



of the wheel is 0.1 m, find the angular acceleration (in rad/sec^2) of the flywheel :-

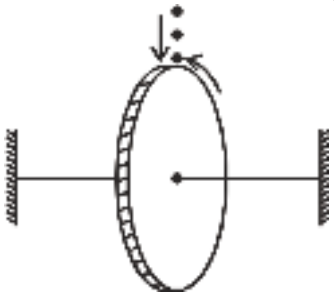
(1) 163.3
(2) 16.66
(3) 81.66
(4) 8.16

16) A disc of mass m and radius R moving in x-y plane as shown. Angular momentum of the disc w.r.t. origin is:



(1) $\frac{5}{2}mR^2\omega$
(2) $\frac{7}{2}mR^2\omega$
(3) $\frac{9}{2}mR^2\omega$
(4) $\frac{3}{2}mR^2\omega$

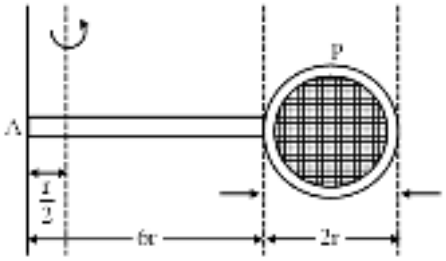
17) A disc of mass 100 gm and radius 25 cm rotates freely about a fixed horizontal axis through its centre. A thin cotton pad is fixed to its rim, which can absorb water.



The mass of water dripping onto the pad is 2 gm per second. After what time will the angular velocity of the disc get reduced to half of its initial value ?

(1) 50 s
(2) 5 s
(3) 10 s
(4) 25 s

18) Consider a badminton racket with length scales as shown in the figure.



If the mass of the linear and circular portions of the badminton racket are same (M) and the mass of the threads are negligible, the moment of inertia of the racket about an axis perpendicular to the handle and in the plane of the ring at, $\frac{r}{2}$ distance from the end A of the handle will be :-

(1) $48mr^2$

(2) $50mr^2$

(3) $52mr^2$

(4) $55mr^2$

19) Two spherical soap bubbles coalesce. If V is the consequent change in volume of the contained air and S is the change in the total surface area and T is the surface tension of the soap solution, then: (if P_0 is atmospheric pressure) :- (temperature does not change)

(1) $3P_0V + 4ST = 0$

(2) $4P_0V + 4ST = 0$

(3) $P_0V + 4ST = 0$

(4) $4P_0V + ST = 0$

20) Two solids A and B float in water. It is observed that A floats with half its volume immersed and B floats with 2/3 of its volume immersed. Compare the densities of A and B:

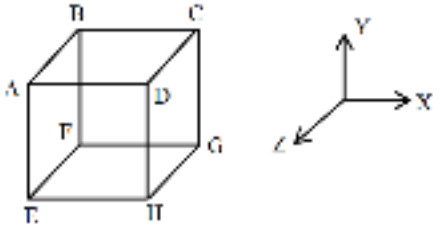
(1) 4 : 3

(2) 2 : 3

(3) 3 : 4

(4) 1 : 3

21) The cubical container ABCDEFGH which is completely filled with an ideal (nonviscous and incompressible) fluid, moves in a gravity free space with acceleration of $\vec{a} = a_0\left(\hat{i} - \hat{j} + \hat{k}\right)$, where a^0 is a positive constant. Then the only point in the container where pressure is maximum, is :-

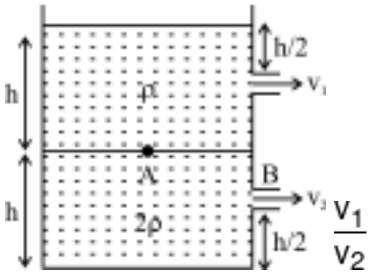


(1) B

(2) C

(3) E

(4) F

22) 

$\frac{v_1}{v_2}$ =?

(1) $\frac{1}{2}$

(2) 1

(3) $\frac{1}{\sqrt{2}}$

(4) $\frac{\sqrt{2}}{\sqrt{3}}$

23) If time taken by water to get empty from a cylinder is 't' s, where the water is filled upto height h, then what is the time taken by water to get empty if it is filled upto height 4h ? The cylinder has a small hole at the bottom of the side wall.

(1) t

(2) 4t

(3) 2t

(4) t/4

24) Two particles of mass 2 kg and 1 kg are moving along the same straight line with speeds 2 m/s and 5 m/s respectively. The speed of COM of system when particles are moving in opposite directions is V_1 and when particles are moving in same direction is V_2. The ratio V_1/V_2 is :

(1) 3:4

(2) 4:3

(3) 9:1

(4) 1:9

25) A uniform thin rod AB of length L has linear mass density $\mu(x) = a + \frac{bx}{L}$, where x is measured from A. If centre of mass of rod lies at A distance of $\frac{7L}{12}$ from A, then a and b are related as :

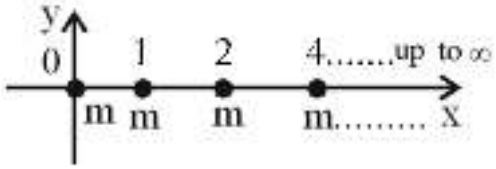
(1) a = 2b

(2) 3a = 2b

(3) 2a = b

(4) a = b

26) A large number of identical point masses m are placed along x-axis, at x = 0, 1, 2, 4,..... The magnitude of gravitational force on mass at origin (x = 0), will be :-



(1) Gm^2

(2) $\frac{4}{3}Gm^2$

(3) $\frac{2}{3}Gm^2$

(4) $\frac{5}{4}Gm^2$

27) If the gravitational field in the space is given as $\left(-\frac{K}{r^2}\right)$. Taking the reference point to be at r = 2 cm with gravitational potential V = 10 J/kg. Find the gravitational potential at r = 3 cm in SI unit (Given, that K = 6 J cm/kg)

(1) 9

(2) 11

(3) 12

(4) 10

28) A planet of mass m move along an ellipse around the sun of mass M, so that its maximum and minimum distances from sun are r_1 and r_2 respectively then angular momentum of this planet relative to the centre of sun is :-

(1) $m\sqrt{\frac{2GMr_1r_2}{(r_1 + r_2)}}$

(2) $m\sqrt{\frac{GMr_1r_2}{(r_1 + r_2)}}$

(3) $m\sqrt{\frac{2GM(r_1 + r_2)}{r_1^2}}$

(4) $m\sqrt{\frac{2GM(r_1 + r_2)}{r_2^2}}$

29) In experiment to calculate coefficient of viscosity by measuring its terminal velocity, it was observed that a ball of mass $\left(\frac{\pi}{2}\right)$ gm and having radius 0.5 cm takes 5 second to fall steadily through a height of 50 cm inside a long column of liquid of density 1.2 gm/cc. The coefficient of viscosity will be given as :

(1) 4.9 poise

(2) 9.8 poise

(3) 12.6 poise

(4) 16 poise

30) A steel wire of diameter 0.4 mm and Young's modulus 2×10^{11} Nm^{-2}, carries a load of mass M. The length of the wire with the load is 1.0 m. A vernier scale with 10 divisions is attached to the end of the wire. Next to the steel wire, is a reference wire to which a main scale of least count 1.0 mm is attached. The 10 division of vernier scale correspond to 9 divisions of the main scale. Initially the zero of the vernier coincides with zero of the main scale. If the load on the wire is increased by 1.5 kg, the vernier scale division which coincides with the main scale division is [Take : g = 10 ms^{-2}, and $\pi \approx 3.2$]

(1) 3rd

(2) 4th

(3) 5th

(4) 6th

31) An asteroid of mass m is approaching towards earth, initially it is at a distance 10 R_E from centre of the earth and moving with speed V_i. It hits earth with a speed V_f (R_E and M_E are radius and mass of earth respectively), then :

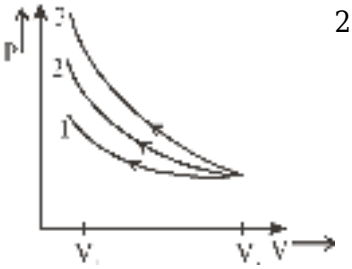
(1) $V_f^2 = V_i^2 + \frac{2Gm}{R_E}\left(1 + \frac{1}{10}\right)$

(2) $V_f^2 = V_i^2 + \frac{2GM_E}{R_E}\left(1 + \frac{1}{10}\right)$

(3) $V_f^2 = V_i^2 + \frac{2GM_E}{R_E}\left(1 - \frac{1}{10}\right)$

(4) $V_f^2 = V_i^2 + \frac{2Gm}{R_E}\left(1 - \frac{1}{10}\right)$

32) P-V plots for three different gases during adiabatic compression are shown in the figure. Plots 1, 2 and 3 should correspond respectively to :-



(1) Monoatomic, diatomic, Polyatomic

(2) Polyatomic, diatomic, Monoatomic

(3) Diatomic, Monoatomic, Polyatomic

(4) Monoatomic, Polyatomic, diatomic

33) 70 calories of heat required to raise the temperature of 2 mole of an ideal gas at constant pressure from 30°C to 35°C. The amount of heat required to raise the temperature of same gas through the same range at constant volume is -

(1) 30 cal

(2) 50 cal

(3) 70 cal

(4) 90 cal

34) If on heating liquid through 80°C, the liquid expelled is $\left(\frac{1}{100}\right)^{th}$ of liquid still remaining in vessel the coefficient of apparent expansion of liquid is :

(1) 1.25×10^{-4}/°C

(2) 12.5×10^{-4}/°C

(3) 1.25×10^{-5}/°C

(4) 1.25×10^{-6}/°C

35) ***Assertion :*** A body that is a good emitter is also a good absorber of radiation at a given wavelength and at a given temperature. ***Reason :*** According to Kirchhoff's law the absorptivity of a body is proportional to its emissivity at a given wavelength and at a given temperature.

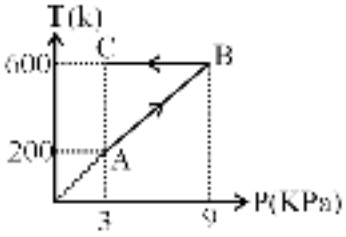
(1) Both assertion & reason are true and reason is the correct explanation of assertion.

(2) Both assertion & reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true & reason is false.

(4) Both assertion & reason are false.

36) 2 moles of a gas is taken from state A to C as shown in given P-T diagram then work done will be.



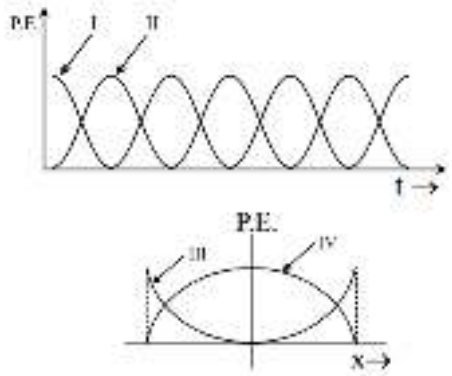
(1) 1200 R (ℓn 3)

(2) 900 R (ℓn 3)

(3) $\frac{400\ R}{\left(\ell n\frac{1}{3}\right)}$

(4) $\frac{1200\ R}{\left(\ell n\frac{1}{3}\right)}$

37) A particle is executing SHM according to x = a cos ωt. Then which of the graphs represents variations of potential energy as a function of time(t) and displacement (x):-



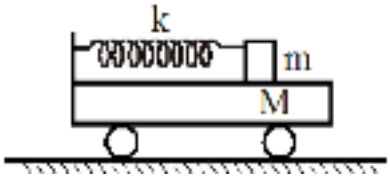
(1) (I) & (III)

(2) (II) & (IV)

(3) (I) & (IV)

(4) (II) & (III)

38) Initially spring is stretched by x_0 and released Find time period of oscillation.



(1) $2\pi\sqrt{\frac{M}{k}}$

(2) $2\pi\sqrt{\frac{M+m}{k}}$

(3) $2\pi\sqrt{\frac{mM}{k(M+m)}}$

(4) $2\pi\sqrt{\frac{M-m}{k}}$

39) Two simple pendulum of length ℓ and 16 ℓ are released from the same phase together. They will be at the same phase after a minimum time :-

(1) $\frac{8\pi}{3}\sqrt{\frac{\ell}{g}}$

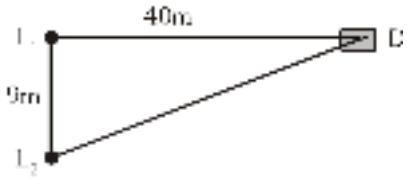
(2) $\frac{\pi}{3}\sqrt{\frac{\ell}{g}}$

(3) 2 s

(4) none of these

40) Two loudspeakers L_1 and L_2, driven by a common oscillator and amplifier, are arranged as shown in fig. The frequency of the oscillator is gradually increased from zero and the detector at D records a series of maxima and minima. If speed of sound is 330 m/s, then the frequency at which the first

maximum is observed is :-



(1) 165 Hz

(2) 330 Hz

(3) 495 Hz

(4) 660 Hz

41) The density of the material of a wire used in sonometer is 7.5×10^5 kg/m^3. If the stress on the wire is 3.0×10^8 N/m^2, the speed of transverse wave in the wire will be :-

(1) 100 m/s

(2) 20 m/s

(3) 300 m/s

(4) 400 m/s

42) Which of the following wave functions does not represent a travelling wave ?

(1) $y = (x - vt)^2$

(2) $y = \log(x + vt)$

(3) $y = \frac{1}{x + vt}$

(4) All of these

43) The velocity of sound is measured by using resonance tube method. The table shows a part of observations mode during the experiment:

Frequency of tuning fork (Hz)	Resonance	Position of water level at resonance L_2 (cm)			Mean length of air column $L_2 - L_1$ (cm)
		Water falling	Water rising	Mean L_2	
512	First	15.8	16.4		□$_1$
	second	49.9	50.3		□$_2$

Room temperature : T = 25° C

Position of upper end of tube = L_1 = 0.2 cm

The speed of sound and the end correction obtained from above data are :

(1) 330 ms^{-1}, 2.2 cm

(2) 348 ms^{-1}, 1.1 cm

(3) 332 ms^{-1}, 0.75 cm

(4) 330 ms^{-1}, 0.75 cm

44) A liquid of specific heat 0.8 cal/g°C at temperature 60°C is mixed with other liquid of same mass having temperature 45°C. If temperature of mixture is 53°C, then specific heat (in cal/g°C) of the liquid is :-

(1) 0.5

(2) 0.6

(3) 0.7

(4) 0.8

45) Three quantities x, y and z are related as $\frac{1}{z} = \frac{1}{x} + \frac{1}{y}$. If x= (100.0 ± 3.0) cm and y = (200.0 ± 4) cm then z will be :

(1) 66.7 ± 1.8 cm

(2) 66.7 ± 3.2 cm

(3) 66.7 ± 0.2 cm

(4) 66.7 ± 2.9 cm

CHEMISTRY

1) **Assertion :-** Friedel craft acylation of benzene is done using acetic anhydride in presence of anhydrous $AlCl_3$.
Reason :- Friedel craft acylation is an example of electrophilic substitution reaction.

(1) Assertion & reason both are correct.

(2) Assertion & reason both are false.

(3) Assertion is correct but reason is false.

(4) Reason is correct but assertion is false.

2) Colour of $[Fe(CN)_5NOS]^{4-}$ is

(1) Blood red

(2) Violet

(3) Green

(4) Yellow

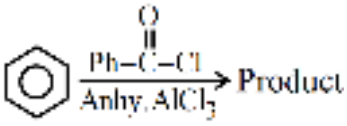
3) In carius method for estimation of halogen, 0.8 g of an organic compound gave 0.94 g of AgBr the percentage of bromine present in the organic compound is -
(Atomic mass of Ag = 108 u and Br = 80 u)

(1) 40%

(2) 50%

(3) 90%

(4) 80%

4) 

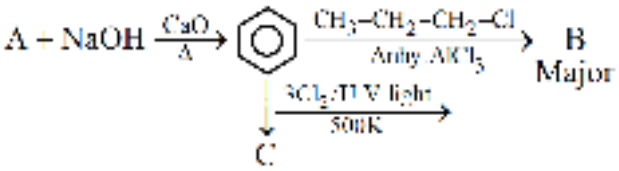
Number of σ and π bonds in product are -

(1) 22 & 6

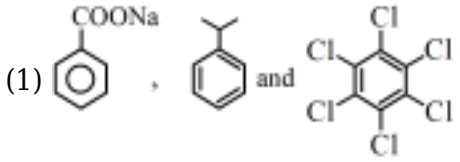
(2) 23 & 6

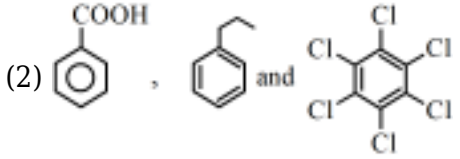
(3) 25 & 7

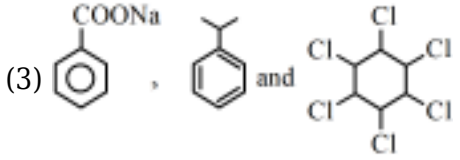
(4) 14 & 4

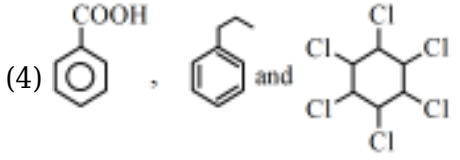
5) 

Compound A, B & C are -

(1) 

(2) 

(3) 

(4) 

6) Out of the given species how many are aromatic in nature ? 

(1) 5

(2) 2

(3) 4

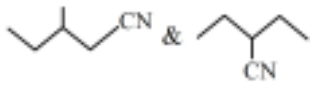
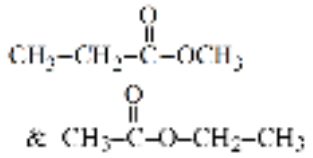
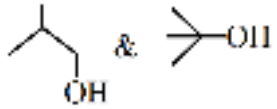
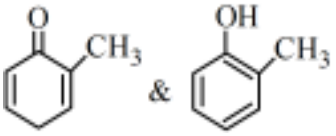
(4) 3

7) Consider the following statements about column chromatography.
(a) Silica gel can be used as an adsorbent in column chromatography.
(b) Thin layer chromatography is a type of adsorption chromatography.
(c) In paper chromatography, the chromatography paper contains water molecules trapped in it

The correct statement are -

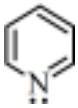
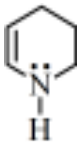
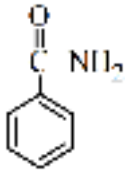
(1) a & b only
(2) b & c only
(3) a & c only
(4) a, b & c

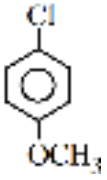
8) Match the column

(A)		(p)	Tautomers
(B)		(q)	Position isomers
(C)		(r)	Metamers
(D)		(s)	Chain isomers

(1) A → s, B → r, C → q, D → p
(2) A → r, B → s, C → q, D → p
(3) A → s, B → r, C → p, D → q
(4) A → p, B → r, C → q, D → s

9) Which of the following is weakest base ?

(1) 
(2) 
(3) 
(4) $Ph–CH_2–NH_2$

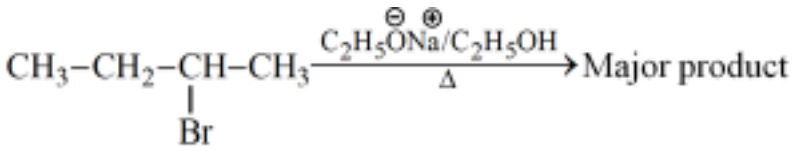

10) IUPAC name of compound is -

(1) 1-chloro anisole

(2) 4-chloro anisole

(3) 4-methoxy chloro benzene

(4) methyl-4-chloro phenoxide ion

11) 

(1) Sec-Butylmethylether

(2) Cis-But-2-ene

(3) Trans-But-2-ene

(4) But-1-ene

12) Find incorrect statement.

(1) Sodium acetate on heating with soda-lime gives methane as major product.

(2) Solvent used in wurtz reaction is dry ether

(3) Ethane has infinite number of conformational isomers

(4) Boiling point of trans But-2-ene is greater than that of cis-But-2-ene

13) Which of the following acts as electrophile in sulphonation in benzene

(1) H_2SO_4

(2) $H_2S_2O_7$

(3) SO_2

(4) SO_3

14) Find correct statement -

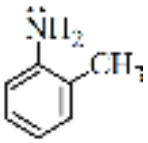
(1) The change in hybridisation, affects the electronegativity of carbon atom

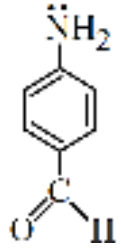
(2) Tetrahydrofuran is hetero cyclic compound

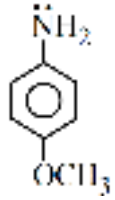
(3) The structural unit $(CH_3)_3C.CH_2$ – is a neopentyl group.

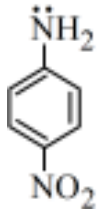
(4) All of these

15) Which of the following is strongest base?

(1) 

(2) 

(3) 

(4) 

16) Select incorrect statement :-

(1) Only three quantum numbers n, ℓ and m are needed to define an orbital

(2) Four quantum numbers are needed for complete description of an electron

(3) Two quantum numbers n and ℓ are needed to identify subshell and shape of orbital

(4) Splitting of spectrum lines in presence of electric field is known as Zeeman effect

17) For the dissociation reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, the degree of dissociation (α) in terms of K_p and total equilibrium pressure P is :-

(1) $\alpha = \sqrt{\frac{P + K_p}{K_p}}$

(2) $\alpha = \sqrt{\frac{K_p}{4P + K_p}}$

(3) $\alpha = \sqrt{\frac{K_p}{4P}}$

(4) $\alpha = \sqrt{\frac{4P + K_p}{K_p}}$

18) Calculate the number of neutrons present in 18 mL of water :- (N_A = Avogadro's number)

(1) 10 N_A

(2) 4 N_A

(3) 8 N_A

(4) N_A

19) It is impossible for a reaction to take place if :-

(1) ΔH is +ve & ΔS is +ve

(2) ΔH is –ve & ΔS is +ve

(3) ΔH is +ve & ΔS is –ve

(4) ΔH is –ve & ΔS is –ve

20) How many H^+ ions are present in 1 mL of a solution whose pH is 12 ?

(1) 10^{-15}

(2) 6.022×10^{8}

(3) 6.022×10^{11}

(4) 10^{7}

21) The order of increasing oxidation number of S in S_8, $S_2O_8^{-2}$, $S_2O_3^{-2}$, $S_4O_6^{-2}$ is –

(1) $S_8 < S_2O_8^{-2} < S_2O_3^{-2} < S_4O_6^{-2}$

(2) $S_2O_8^{-2} < S_2O_3^{-2} < S_4O_6^{-2} < S_8$

(3) $S_2O_8^{-2} < S_8 < S_4O_6^{-2} < S_2O_3^{-2}$

(4) $S_8 < S_2O_3^{-2} < S_4O_6^{-2} < S_2O_8^{-2}$

22) **Statement-I :-** Actual mass of one molecule of CH_4 is 16 amu.
Statement-II :- Mass percentage of carbon in CH_4 is 80%.

(1) Both statements are correct

(2) Statement-I is correct, statement-II is incorrect

(3) Statement-II is correct, statement-I is incorrect

(4) Both statement are wrong

23) $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$; K = 5
How many moles of CO_2 must be added to 1 L container already containing 3 moles each of CO and H_2O to make 2M equilibrium concentration of CO :-

(1) 15

(2) 19

(3) 5

(4) 20

24)

The entropy change for the reaction
$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$ will be
[Given that, $S°(HCl) = 187\ JK^{-1}\ mol^{-1}$]
$S°(H_2) = 131\ JK^{-1}\ mol^{-1}$ and
$S°(Cl_2) = 223\ JK^{-1}\ mol^{-1}$

(1) $20\ JK^{-1}\ mol^{-1}$

(2) $-20\ JK^{-1}\ mol^{-1}$

(3) 167 JK^{-1} mol^{-1}

(4) – 167 JK^{-1} mol^{-1}

25) **Assertion (A) :** Oxygen atom in both, O_2 & O_3 has an oxidation number 0.
Reason (R) : Oxygen is assigned an oxidation number –2 in all of its compounds.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is not correct

(4) (A) and (R) both are incorrect

26)

The degree of hydrolysis of a salt of weak monobasic acid and weak monoacidic base in its 0.1 M solution is found to be 50%. If the molarity of the solution is 0.2 M, the percentage hydrolysis of the salt should be :

(1) 100%

(2) 50%

(3) 25%

(4) None of these

27) Find the ratio of the time period of 2nd Bohr orbit of He^{+} ion and 4th Bohr orbit of Li^{2+} ion.

(1) 32/9

(2) 9/16

(3) 9/32

(4) 16/9

28) The number of moles of oxalate ions oxidized by one mole of MnO_4^- ion in acidic medium.

(1) $\frac{5}{2}$

(2) $\frac{2}{5}$

(3) $\frac{3}{5}$

(4) $\frac{5}{3}$

29) pH of 0.01 M of a monobasic weak organic acid is 5.10 calculate K_a value of acid :

(1) 1.6×10^{-7}

(2) 6.4×10^{-9}

(3) 10^{-8}

(4) 6.4×10^{-7}

30) In the beginning of the reaction, A ⇌ B + C, 2 moles of A are taken, out of which 0.5 moles gets dissociated. What is the degree of dissociation of A ?

(1) 0.5

(2) 1

(3) 0.25

(4) 4.2

31)

200 gm $CaCO_3$(s) is heated and it decomposed into CaO(s) and CO_2(g). What is the weight of residue -(Atomic weight of Ca = 40)

(1) 56 gm

(2) 200 gm

(3) 100 gm

(4) 112 gm

32) Given below are two statements:
Statement-I : The energy of He^+ ion in n = 2 state is same as the energy of H atom in n = 1 state.
Statement-II : It is possible to determine simultaneously the exact position and exact momentum of an electron in H atom.
In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **Statement I** and **Statement II** are false.

(2) **Statement I** is true but **Statement II** is false.

(3) **Statement I** is false but **Statement II** is true.

(4) Both **Statement I** and **Statement II** are true.

33) Following reaction is occuring in an automobile engine
$2C_8H_{18(g)} + 25O_{2(g)} \rightarrow 16\ CO_{2(g)} + 18\ H_2O_{(g)}$
The sign of ΔH, ΔS and ΔG would be :-

(1) +, – , +

(2) –, +, –

(3) –, +, +

(4) +, +, –

34) For reaction aA ⇌ □L + mM. In condition of sudden volume increase, degree of dissociation decreases it represents that.

(1) a < (□ + m)

(2) a = (□+ m)

(3) a = (□ – m)

(4) a > (□ + m)

35) Which of the following is most soluble at same temperature in water :-

(1) MnS ($K_{SP} = 10^{-16}$)

(2) CuS ($K_{SP} = 10^{-36}$)

(3) ZnS ($K_{SP} = 10^{-24}$)

(4) FeS ($K_{SP} = 10^{-18}$)

36) A ⇒ [Ar] $3d^5 4s^1$ B ⇒ [Ar] $3d^5 4s^2$
C ⇒ [Ar] $3d^{10} 4s^2 4p^1$ D ⇒ [He] $2s^2 2p^4$
Which of the following compound is considered to be as an amphoteric compound

(1) AD_3 , C_2D

(2) BD, AD

(3) C_2D_3 , A_2D_3

(4) AD, B_2D_7

37)

Among the following species identify the isostructural pairs:-
NF_3, NO_3^-,BF_3, H_3O^+, HN_3

(1) [NF_3, NO_3^-] and [BF_3, H_3O^+]

(2) [NF_3, HN_3] and [H_3O^+,BF_3]

(3) [NF_3, H_3O^+] and [NO_3^- , BF_3]

(4) [NF_3, H_3O^+] and [HN_3, BF_3]

38) Which of the following order is incorrect :-

(1) $PCl_3F_2 < PF_3Cl_2$ Dipole moment

(2) N–N < P–P bond length

(3) $PH_3 < NH_3$ % s in l.p.

(4) $PH_3 < NH_3$ bond angle

39) Which of the following is not correctly matched ?

1.	SF_4	Folded square shape
2.	XeF_3^+	Bent - T - shape
3.	XeO_6^{4-}	Square pyramidal
4.	PCl_6^-	Square bipyramidal

(1) 1

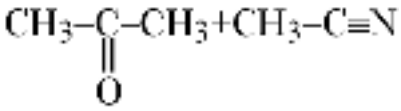
(2) 2

(3) 3

(4) 4

40) Dipole-dipole attractive (keesome) is present in.

(1) $KCl + H_2O$

(2) 

(3) $NO_3^- + Cl_2$

(4) $HCl + Cl_2$

41) In which of the following species the bonds are non-directional?

(1) NCl_3

(2) RbCl

(3) $BeCl_2$

(4) BCl_3

42) Which of the following order of ionization energy is correct :-

(1) $K^+ > Cu^+ > Cu > K$

(2) $Cu^+ > K^+ > Cu > K$

(3) $K^+ > Cu^+ > K > Cu$

(4) $Cu^+ > K^+ > K > Cu$

43) Consider the following electronic configuration of an element (P) : $[Xe]4f^{14}5d^16s^2$
Statement-I : It belongs to 6th period and 3rd group.
Statement-II : It belongs to actinoid series.

(1) Statement I and Statement II are true.

(2) Statement I and Statement II are false.

(3) Statement-I is true and statement-II is false

(4) Statement-I is false and statement-II is true

44) Given below are two statements :
Statement-I : The hybridised orbital are generally equivalent in energy and shape
Statement-II : The hybrid orbital are effective in forming stable bonds than the pure atomic orbitals.

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct and statement-II is incorrect

(4) Statement-I is incorrect and statement-II is correct

45) ***Assertion :-*** Dipole moment of CH_4 and BF_3 are zero.
Reason :- Both have same geometry.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

BIOLOGY

1) The fundamental element of taxonomy which includes the grouping of organism into convenient categories on the basis of easily observable characters is :-

(1) Identification
(2) Nomenclature
(3) Classification
(4) All of the above

2) Animals, mammals and dogs represents :-

(1) Taxa at same level
(2) Taxa at different level
(3) Group of same species
(4) same class

3) Which one of the following statement is incorrect ?

(1) Classification is not a single step process but involves hierarchy of steps in which each step represents a rank or category.
(2) The taxonomic groups/categories are distinct biological entities and not merely morphological aggregates.
(3) Genus comprises a group of related species which has more characters in common in comparison to species of other genera.
(4) As we go higher from species to kingdom the number of common characteristics goes on increasing.

4) Red tide is caused by :-

(1) Excessive growth of a red alga
(2) Excessive growth of a blue green alga
(3) Excessive growth of a protist
(4) Excessive growth of a red bacterium

5) Which of the following statement is correct?

(1) The mycelium is branched and septate in basidiomycetes.
(2) Ascospores are produced exogenously
(3) The mycelium is septate and uninucleate in *Saccharomyces*.
(4) Both sexual and asexual phase occur in deuteromycetes.

6) According to five kingdom system scheme, which statement is common about given organism :- *Euglena, Nostoc, Chlorella,* Diatoms and Dinoflagellates

(1) All are unicellular eukaryotes

(2) All are multicellular eukaryotic organism

(3) All are unicellular prokaryotes

(4) They are photosynthetic and act as producer in nature

7) Match the column I, II & III

Column I		**Column II**		**Column III**	
1	*Albugo*	(a)	Ascomycetes	(i)	Obligate parasite
2	*Agaricus*	(b)	Deuteromycetes	(ii)	Decomposer of litter
3	*Aspergillus*	(c)	Phycomycetes	(iii)	Branched and septate mycelium
4	*Alternaria*	(d)	Basidiomycetes	(iv)	edible fungus

(1) 1-c-i, 2-a-ii, 3-a-iii, 4-b-iv

(2) 1-c-i, 2-d-iv, 3-d-ii, 4-a-iii

(3) 1-c-i, 2-d-iii, 3-d-iv, 4-b-ii

(4) 1-c-i, 2-d-iv, 3-a-iii, 4-b-ii

8) List of some organisms is given below :-
Lichen, Virus, Bacteria, *Riccia*, *Chlorella*, *Chlamydomonas*, *Capsella*, *Euglena*, *Gonyaulax*, *Chara*
According to R.H. whittakar how many belongs to kingdom protista?

(1) Three

(2) Four

(3) Five

(4) Six

9) Identify the correct statements :
I – *Paramoecium* has cellular body organisation with heterotrophic mode of nutrition.
II – Cell structure, body organisation, mode of nutrition reproduction and phylogenetic relationships were the main criteria of classification used in five kingdom classification.
III – The cell wall of eubacteria is made up of peptidoglycan.
IV – *Neurospora* and *Ustilago* shows saprophytic mode of nutrition and have cell wall made up of cellulose.
V – The nucleus of *Nostoc* and *Anabaena* is membrane bound.
Choose the correct answer from the options given below.

(1) I, II and III only

(2) II and V only

(3) II and III only

(4) III, IV and V only

10) Gametes of Phaeophyceae are

(1) Comma shaped uniflagellated

(2) Spiral shaped Multiflagellated

(3) pear shaped Biciliated

(4) Pyriform Biflagellated

11) What is common between gymnosperm and angiosperm ?

(1) Fruit formation

(2) Xylem and phloem present

(3) Seed formation

(4) Both (2) and (3)

12) **Assertion :** In gymnosperms, bisexual cones are developed on same tree.
Reason : In gymnosperms, cones are developed by spirally arranged sporophylls.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is False but the Reason is True.

(4) Both Assertion & Reason are False.

13) **Statement-I :** Majority of Pteridophytes produces two kind of spore (macro & micro) and are known as heterosporus.
Statement-II : Development of Zygotes into young embryo takes places within female gametophyte in homosporus pteridophytes.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II are incorrect.

14) Which of the following is not correctely matched ?

(1) *Chlamydomonas* – Unicellular flagellated

(2) *Laminaria* – Flattened leaf-like thallus

(3) *Chlorella* – Unicellular non-flagellated

(4) *Volvox* – Colonial form, non-flagellated

15) Which of the following require water for fertilization?

(A)  (B) 

(C)  (D) 

(1) A and B

(2) A and C

(3) A and D

(4) A, B and C

16) A. Cytoskeleton is proteinaceous and help in motility and cell division.
B. Nucleoplasm contains nucleolus and chromatin.
C. A single somatic human cell has approximately one meter long thread of DNA.

(1) A, C are correct

(2) B, C are correct

(3) A, B are correct

(4) All are correct

17) Glycogen synthesis in animal cell occurs in :

(1) ER

(2) Lysosomes

(3) Mitochondria

(4) Golgi body

18) Plasma membrane is :

(1) Selectively permeable and made up of cellulose.

(2) Semi permeable and made up of lipid and carbohydrate mainly.

(3) Permeable and made up of phospholipid and protein.

(4) Selectively permeable and made of phospholipid and protein.

19) Which of the following statement is incorrect?

(1) Lysosomal enzymes are capable of digesting carbohydrates, proteins, lipids and nucleic acids

(2) In protists food vacuoles are formed by engulfing the food particles

(3) Cis and trans faces of the golgi apparatus are entirely different, but are interconnected

(4) Tonoplast facilitates the transport of a number of ions and other materials along concentration gradients into the vacuole

20) Select the option which is correctly matched.

(i) Single chromatid chromosome - Prophase
(ii) Two chromatid chromosome - Metaphase
(iii) Two chromatid chromosome - Anaphase
(iv) NOR - Secondary constriction
(v) Nucleolus - Production of ribosome

(1) (ii), (iv), (v)
(2) (i), (iii), (iv)
(3) (ii), (iii), (iv), (v)
(4) (i), (v)

21) How many chromosome will the somatic cell have during mitosis at G_1, after S and in anaphase respectively if sperm of this organism has seven chromosomes :-

(1) 14, 28, 28
(2) 14, 14, 28
(3) 7, 7, 7
(4) 7, 14, 14

22) Spindle fibres attach on to :-

(1) Telomere of the chromosome
(2) Kinetochore of the chromosome
(3) Centromere of the chromosome
(4) Secondary constriction of the chromosome

23) Arrange the following events of meiosis in the correct sequence :-
(A) Spliting of the centromere
(B) Terminalisation of the chiasmata
(C) Synapsis
(D) Diad of the cell
(E) Dissolution of synaptonemal complex
(F) Crossing over

(1) E → C → B → D → A → F
(2) C → E → F → B → D → A
(3) C → F → E → B → A → D
(4) C → F → E → B → D → A

24)

Which of the following statements about meiosis is false ?

(1) Haploid cells can not divide meiotically
(2) DNA replication does not occur during interkinesis
(3) Assortment of homologous chromosomes on the equatorial plate during metaphase-I is independent

(4) Of the four cells produced by meiosis, none are genetically identical to the parent cells but two are identical to each other.

25) **Statement-I** : Metaphase is a second phase of M-phase in which number of chromosomes and amount of DNA remain same as prophase.
Statement-II : The number of chromosomes become double in anaphase as compare to metaphase and the amount of DNA remain same as prophase.
Select the correct option :

(1) Both statement-I and statement-II are correct
(2) Both statement-I and statement-II are incorrect
(3) Only statement-I incorrect
(4) Only statement-II incorrect

26)

How many carbon atoms are found in palmitic acid ?

(1) 12
(2) 14
(3) 16
(4) 18

27) Example of alkaloid and lacitin respectively are :-

(1) Carotenold and morphine
(2) Codeine and concanavalin A
(3) Codine and lemon gran oil
(4) Vinblastin and curcumin

28)

Select the correct option :
Statement I - Tertiary Structure is absolutely necessary for the many biological activities of protein
Statement II - Adult human haemoglobin consist of 4 subunits.

(1) Both **Statement I** and **Statement II** are correct.
(2) Both **Statement I** and **Statement II** are incorrect.
(3) **Statement I** is correct but **Statement II** is incorrect.
(4) **Statement II** is correct but **Statement I** is incorrect

29) Read the following statements and select the correct option :-
(A) Right end of the polysaccharide chain is called reducing end while left end is called non reducing end.
(B) Starch can hold iodine molecule in its helical secondary structure but cellulose being non helical, can not hold iodine.
(C) Starch and glycogen are branched molecules.

(D) Starch and glycogen are reserve food materials of plants and animals respectively.

(1) Statement A and B are correct.

(2) Statement B and C are correct.

(3) Only statement D is correct.

(4) All statements A, B, C and D are correct.

30) From the statements given below choose the correct option :
(A) Collagen is the most abundant protein in animal world.
(B) Complex polysaccharides are mostly homopolymers.
(C) Protein is a hetero polymer not a homopolymer.
(D) Inulin is a homopolymer of glucose.
(E) Glycine is not a sulphur containing amino acid.

(1) A, B, D are true

(2) A, D, E are true

(3) A, B, E are true

(4) B, D, E are true

31) **Statement-I :** A chromatograpic separation of the leaf pigments shows that the colour that we see in leaves is not due to single pigment but due to four pigments : Chlorophyl-a (yellow green), Chlorophyll-b (blue green), Xanthophyll (yellow) and Carotenoids (yellow to yellow organe).
Statement-II : The wavelengths at which there is maximum absorption by chlorophyll-a i.e in blue and red region, also shows higher rate of photosynthesis hence, we can conclude that chlorophyll-'d' is the chief pigment associated photosynthesis.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

32) Consider the following statements about the light reactions of photosynthesis :
(A) Water splitting directly occurs at photosystem I.
(B) The first electron acceptor of photosystem I is plastocyanin.
(C) ATP and NADPH are produced in the light reaction.
(D) Oxygen is released as by product.
Which of the above statements are correct?

(1) A, C, D only

(2) A, B, C only

(3) C, D only

(4) A, B, D only

33) In a classic Avena coleoptile curvature bioassay, a group of students removed the coleoptile tips of oat seedlings and placed small agar blocks on one side of the cut tips. These agar blocks had been kept previously on auxin-producing coleoptile tips, allowing auxin to diffuse into them.
After about 2 hours, the coleoptiles showed a clear bending response - the tip curved away from the side where the auxin-block was placed.

Based on this observation, what can be correctly concluded about the action of auxin ?

(1) Auxin stimulates faster elongation on the side where it is present, leading to bending of the coleoptile.

(2) Auxin inhibits cell elongation on the opposite side, causing curvature.

(3) Auxin has equal effect on both sides but still produces curvature due to uneven agar placement.

(4) Auxin primarily promotes cell division on the treated side, which results in bending.

34)

	Column I (Stage of Aerobic Respiration)		**Column II** (Primary Location in Eukaryotic Cell)		**Column III** (Major Output per Glucose Molecule)
(A)	Glycolysis	(i)	Mitochondrial Matrix	(1)	2 ATP (net), 2 NADH, 2 pyruvate
(B)	Pyruvate Oxidation (Link Reaction)	(ii)	Cytosol	(2)	2 Acetyl-CoA, 2 NADH, $2CO_2$
(C)	Citric Acid Cycle (Krebs cycle)	(iii)	Inner mitochondrial membrane	(3)	6 NADH, 2 $FADH_2$, 2ATP (GTP), $4CO_2$
(D)	Electron Transport Chain & Oxidative Phosphorylation	(iv)	Intermembrane space (buildup utilized)	(4)	~34 ATP

(1) A-(i)-(1); B-(ii)-(2); C-(ii)-(3); D-(iii)-(4)

(2) A-(ii)-(1); B-(i)-(2); C-(i)-(3); D-(iii)/(iv)-(4)

(3) A-(i)-(1); B-(iii)-(3); C-(ii)-(2); D-(iv)-(4)

(4) A-(iv)-(4); B-(ii)-(2); C-(i)-(3); D-(iii)-(1)

35) **Assertion :-** Glycolysis, link reaction, kreb's cycle, ETS occur during cellular respiration in aerobic condition.
Reason :- During cellular respiration, all the energy contained in respiratory substrates is not released into a single step but it is released in a series of slow step wise reactions controlled by various enzymes.

(1) Both Assertion and Reason are incorrect.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(4) Assertion is correct but Reason is incorrect.

36) Acetyl CoA :-

(1) Connects fatty acids to respiratory path

(2) Connect glycerol to respiratory pathway

(3) is the product of Glycolysis

(4) is a 3 carbon compound

37) In fermentation there is a net gain of only _____ molecules of ATP for each molecule of glucose degraded to pyruvic acid. When proteins are respiratory substrate the RQ would be about _____.

(1) 2, 1
(2) 8, 1
(3) 2, 0.9
(4) 8, 0.9

38) How many cytochromes in the list given below are the member of respiratory electron transport system ? Cytochrome d, Cytochrome a, Cytochrome C_1, Cytochrome C_6, Cytochrome a_3, Cytochrome b, Cytochrome d_3

(1) Five
(2) Four
(3) Six
(4) Two

39) Which of the following is referred as cytochrome C oxidase ?

(1) Complex IV
(2) Complex III
(3) Complex II
(4) Complex I

40) Adenine derivative plant growth regulator is

(1) IAA
(2) Kinetin
(3) ABA
(4) C_2H_4

41) In which of the following PGRs promote nutrient mobilisation?

(1) Auxin
(2) Cytokinin
(3) Gibberellins
(4) Ethylene

42) Spraying juvenile confers with which phytohormones hastens the maturity period, thus leading to early seed production :

(1) Auxin
(2) Gibberellins
(3) Cytokinin

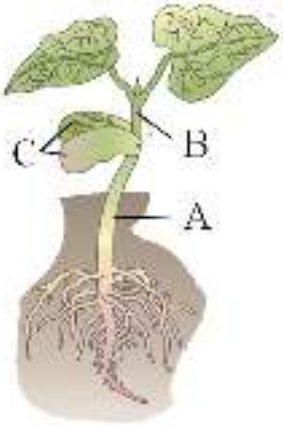
(4) Ethylene

43) Which of the following hormones helps to promotes rapid internode / petiole elongation in deep water rice plant ?

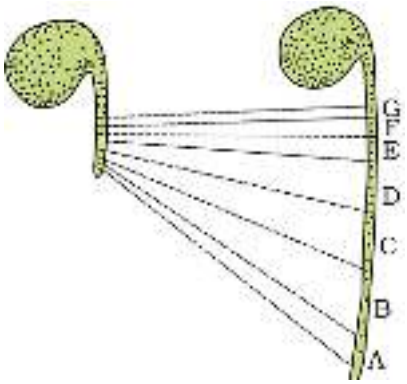
(1) Ethylene
(2) Abscisic acid
(3) Cytokinin
(4) Auxins

44)

The figure shows a diagrammatic view with labelling A, B and C. Select the options which give correct identifications :



(1) A-Cotyledons, B-Hypocotyle, C-Epicotyl
(2) A-Epicotyl, B-Hypocotyl, C-Cotyledons
(3) A-Hypocotyl, B-Epicotyl, C-Cotyledons
(4) A-Hypocotyl, B-Hypocotyl, C-Cotyledons



45) Which of the following is the correct regarding the given diagram?

(1) Helpful in Detection of zones of maturation by the parallel line technique.
(2) Zones A, B, C, D immediately behind the apex have elongated most.
(3) increase in protoplasm is not difficult to measure directly,
(4) plant growth and further development is not intimately linked to the water status of the plant.

46) A viviparous mammal which is found only in Australia:

(1) Pteropus
(2) Canis
(3) Elephas
(4) Macropus

47)

Correct path of flow of water in sponges is

(1) Spongocoel - Ostia - Osculum
(2) Osculum - Ostia - Spongocoel
(3) Spongocoel - Osculum - Ostia
(4) Ostia - Spongocoel - Osculum

48)

Match List-I with List-II and choose the correct option :-

	List-I		**List-II**
(A)	Shed scales as skin cast	(i)	Platypus
(B)	Air sacs associated with lungs	(ii)	Lizzard
(C)	Oviparous mammal	(iii)	Corvus

	A	**B**	**C**
(1)	i	ii	iii
(2)	ii	i	iii
(3)	ii	iii	i
(4)	iii	ii	i

(1) 1
(2) 2
(3) 3
(4) 4

49) Given below are two statements :
Statement-I : Notochord is a mesodermally derived rod-like structure formed on the dorsal side during embryonic development in some animals.
Statement-II : Animals with notochord are called chordates.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

50) Match the column I, II and III & choose the correct answer :-

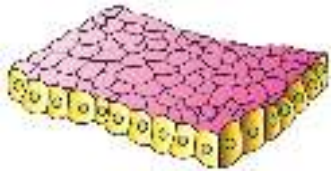
	Column-I Animal	**Column-II Feature**	**Column-III Phylum/Class**
(1)	*Psittacula*	Warm blooded	Hemichordata
(2)	*Myxine*	Nerve cord absent	Cephalochordata
(3)	*Ichthyophis*	Limbs absent	Amphibia
(4)	*Prawn*	Chitinous endoskeleton	Chordata

(1) 1
(2) 2
(3) 3
(4) 4

51) How many of the following characteristics are present in humans ?

Muscular diaphragm	Seven cervical vertebrae	Mammary glands
Body hair	Oviparity	Corpus callosum
Four chambered heart	Direct development	Thecodont dentition
One pair of forelimbs and one pair of hindlimbs		

(1) Seven
(2) Eight
(3) Nine
(4) Ten



52) Identify the given below figure with its correct example

(1) Squamous epithelium – Air sacs of lungs
(2) Cuboidal epithelium – Tubular part of nephron
(3) Columnar epithelium – Fallopian tube
(4) Cuboidal epithelium – Wall of blood vessels

53) Which cell is chief matrix producing cells ?

(1) Macrophages cell
(2) Mast-cells
(3) Fibroblast cell
(4) Adipose cells

54) In cockroach choose incorrect statement(s)
(a) Tergo-sternal muscles help in excretion
(b) Nerve cord has seven segmental ganglia in abdomen
(c) long tubule of utricular gland help in the formation of middle layer of spermatophore

(1) a,b and c
(2) only a and b
(3) only a and c
(4) only b and c

55) Which of the following structures forms the lateral wall of the Pre-oral cavity ?

(1) Labrum, Mandible
(2) Maxilla, Labium
(3) Mandible, Maxilla
(4) Maxilla, Hypopharynx

56) The nerve cord of Cockroach is–

(1) Double, Ventral, Solid
(2) Double, Dorsal, Hollow
(3) Single, Dorsal, Solid
(4) Single, Ventral, Solid

57) A mature female frog can lay how many ova at a time ?

(1) 1500 to 2000
(2) 2000 to 2500
(3) 2500 to 3000
(4) 500 to 1000

58) Which type of respiration occurs in frog during hibernation and aestivation ?

(1) Both cutaneous and pulmonary
(2) Both cutaneous and buccopharyngeal
(3) Only cutaneous
(4) Only pulmonary

59) Correct statement related to frog is :-

(1) Control and coordination system is highly evolved in frog consisting of both neural and endocrine system.
(2) Endocrine glands in frog include pituitary, thyroid and other glands but parathyroid and thymus absent.
(3) Nervous system in frog is non organised having CNS only.
(4) Brain of frog has forebrain and hindbrain only.

60) Carotid artery carries :-

(1) Impure blood to kideneys
(2) Oxygenated blood to brain
(3) Impure blood from brain
(4) Oxygenated blood to heart

61) ***Assertion (A) :*** During diastole, ventricles receives blood from auricles.
Reason (R) : During ventricular diastole (stage of rapid inflow) the pressure inside auricles is greater than ventricles.

(1) Both (A) and (R) are true and (R) is correct explanation of (A)
(2) Assertion and Reason both are true but Reason is not correct explanation of Assertion
(3) Assertion and Reason both are false
(4) Assertion is false while Reason is true

62) Match the column-I to column-II :-

Column-I		**Column-II**	
(I)	Polycythemia	(a)	Increased no. of acidophil
(II)	Leucocytosis	(b)	Decreased no. of platelets
(III)	Thrombocytopenia	(c)	Increased no. of R.B.C
(IV)	Eosinophilia	(d)	Increased T.L.C

(1) (I)-(c) (II)-(b) (III)-(d) (IV)-(a)
(2) (I)-(b) (II)-(a) (III)-(c) (IV)-(d)
(3) (I)-(c) (II)-(d) (III)-(b) (IV)-(a)
(4) (I)-(d) (II)-(a) (III)-(b) (IV)-(c)

63)

What is the correct order of these events :
(A) Conversion of prothrombin to thrombin
(B) Clot formation and leakage of serum
(C) Thromboplastin formation
(D) Conversion of fibrinogen to fibrin

(1) C → B → A → D
(2) C → A → D → B
(3) C → D → B → A
(4) D → A → C → B

64) Read the following statements :-
(a) Systemic circulation takes place between left ventricle and right atrium of heart.
(b) Adrenal medullary hormones can decrease the cardiac output.
(c) Heart failure is not the same as heart attack.

(d) The opening between left atrium and left ventricle is guarded by bicuspid valve.
Out of these how many statements are correct ?

(1) One

(2) Two

(3) Three

(4) Four

65) **Assertion(A) :-** When a graph is plotted between percentage saturation of foetal Hb with O_2 and partial pressure of oxygen the graph should be shifted towards left side as compare to adult Hb.
Reason(R) :- The foetal Hb have more binding affinity towards O_2 as compare to adult haemoglobin hance have less P_{50} value.

(1) Both (A) and (R) are true and (R) is correct explanation of (A)

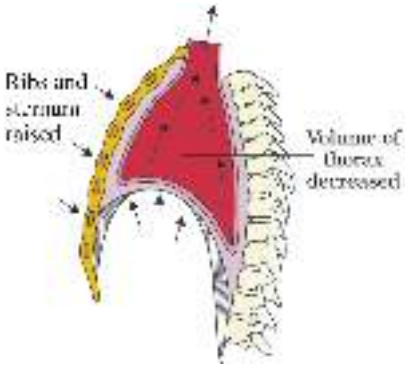
(2) Assertion and Reason both are true but Reason is not correct explanation of Assertion

(3) Assertion and Reason both are false

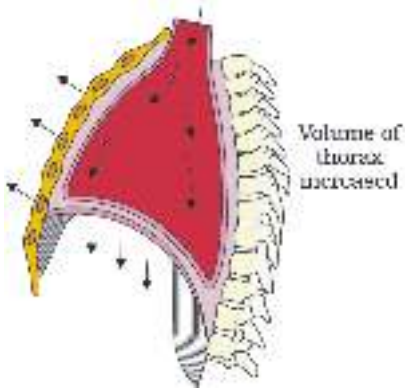
(4) Assertion is false while Reason is true

66) The figure shows mechanism of breathing i.e. (inspiration).
Choose the most appropriate answer from the options given below :

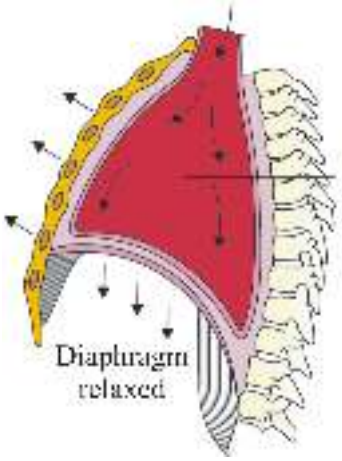
(1)



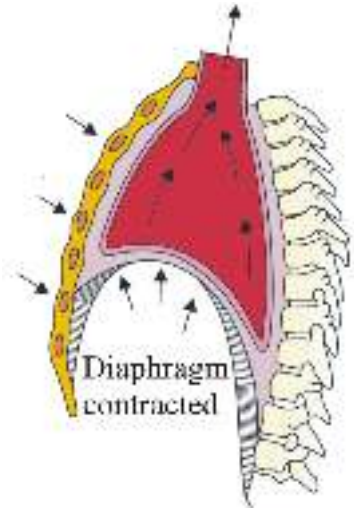
(2)



(3)



(4)



67) Consider the following statements related to the neural regulation of respiration:
A. The respiratory rhythm center is located in the medulla region of the brain.
B. The pneumotaxic center in the medulla increases the duration of expiration.
C. The chemosensitive area adjacent to the rhythm center is highly sensitive to CO_2 and hydrogen ions.
D. The role of oxygen in the regulation of respiratory rhythm is significant.
Which of the above statements are correct?

(1) A and C

(2) B and D

(3) A, C and D

(4) A, B and C

68) The partial pressure of oxygen in the alveoli of the lungs is

(1) equal to that in the blood

(2) more than that in the blood

(3) less than that in the blood

(4) less than that of carbon dioxide

69) C-shaped cartilagenous rings in Human respiratory tract are found ?

(1) From trachea to end of bronchioles

(2) From trachea upto alveoli

(3) Only in trachea and primary bronchi

(4) From trachea to initial bronchioles

70) Read the following terminology of column-I & II identify their correct match -

Column-I		Column-II	
(A)	A cup-like structure	(i)	Collecting duct
(B)	A U-shaped structure	(ii)	PCT
(C)	A straight tube	(iii)	Hilum
(D)	A notch	(iv)	Calyx
		(v)	DCT
		(vi)	Bowman's capsule
		(vii)	Vasa recta

(1) (A)-(vi), (B)-(iii), (C)-(vii), (D)-(iv)

(2) (A)-(iii), (B)-(vi), (C)-(i), (D)-(vii)

(3) (A)-(vi), (B)-(vii), (C)-(i), (D)-(iii)

(4) (A)-(vi), (B)-(v), (C)-(ii), (D)-(iv)

71) **Assertion :** In the descending limb of loop of Henle, urine is hypertonic, while is ascending limb the urine is hypotonic.
Reason : Descending limb is impermeable to Na^+, while ascending limb is impermeable to H_2O.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

72)

You and your study partner want to draw the pathway that controls the reabsorption of sodium ion when blood pressure falls. Which of the following is the correct sequence of the events ?
(i) Aldosterone is released
(ii) Kidney tubules reabsorb Na^+
(iii) Renin is released
(iv) JGA recognises a drop in BP
(v) Angiotensin II is produced.

(1) i–iii–v–ii–iv

(2) iv–ii–iii–i–v

(3) iv–iii–v–i–ii

(4) iv–iii–v–ii–i

73) Identify the incorrect statement from the following

(1) Mammals have the ability to produce a concentrated urine.

(2) Transport of substance is facilated by the special arrangement of henle's loop and vasa Recta.

(3) Counter current mechanis is not involved to maintain a concentration in medullary interstitium

(4) NaCl is transported by the ascending limb of henle's loop which is exchanged with the descending limb of vasa Recta.

74) Dialysis is done in the condition when person is suffering from :

(1) Diabetes

(2) Uraemia

(3) Polyuria

(4) Haemolysis

75) Given below are two statements :-
Statements 1 :- The midbrain is located between the thalamus/hypothalamus of the hindbrain and pons of the forebrain.
Statements 2 :- The ventral part of the midbrain consists mainly of four round swellings called corpora quadrigemina.
Choose the correct answer from the following options :-

(1) Both statement 1 and statement 2 are false.

(2) Statement 1 is true but statement 2 is false.

(3) Statement 1 is false but statement 2 is true.

(4) Both Statement 1 and 2 are true.

76) Which of the following statement is incorrect ?

(1) Electrical synapses are rare in our system.

(2) In depolarisation stage there is a rapid influx of K^+.

(3) Axon terminals contain vesicles filled with neurotransmitters.

(4) Neurons are excitable cells.

77) The ionic gradients accross the resting membrane are maintained by active transport of ions by :-

(1) Sodium voltage gated channels

(2) Potassium voltage gated channels

(3) Ion channels

(4) Sodium potassium pump

78) Cardiovascular reflexes and gastric secretions are controlled by -

(1) Pons

(2) Medulla oblongata

(3) Cerebrum

(4) Spinal cord

79) Read the following statement :-
(a) Hormone are nutrient chemicals.
(b) Hypothalamus is basal part of olfactory lobe
(c) Pancreas is organised endocrine gland.
(d) Posterior pituitary is under direct control of hypothalamus.
(e) G.H is released by anterior pituitary
Choose the incorrect statement :-

(1) a,b

(2) c,e

(3) a,c,d

(4) b,c,d,e

80) How many hormone in given list are releated with immune system :-
Thyroxine, Aldosterone, Thymosin, Glucagon, Growth hormone, Cortisol, Testosterone

(1) Six

(2) Four

(3) Two

(4) One

81) Select the correctly matched pair of the hormones given in column I and their glands in column II

	Column-I (Hormones)		**Column-II (glands)**
a.	Thymosin	i	Gastro-Intestinal Tract
b.	Insulin	ii	Thymus
c.	Adrenaline	iii	Pancreas
d.	Gastrin	iv	Adrenal medulla

(1) a - ii, b - iii, c - i, d - iv

(2) a - ii, b - iii, c - iv, d - i

(3) a - iii, b - i, c - ii, d - iv

(4) a - iv, b - ii, c - iii, d - i

82) Regarding mechanism of FSH hormone, select the correct sequential steps :-
(A) Hormone receptor complex formation
(B) Biochemical-response
(C) Generation of cyclic-Amp
(D) Physiological-response
Choose the correct answer from the options given below :-

(1) A, D, C, B

(2) A, B, C, D

(3) A, C, B, D

(4) A, C, D, B

83) Which one of the following pairs correctly matches a hormone with a disease resulting from its deficiency?

(1) Luteinizing hormone – Failure of ovulation
(2) Insulin – Diabetes insipidus
(3) Thyroxine – Tetany
(4) Parathyroid hormone – Diabetes mellitus

84) Which of the following hormone prevents diuresis?

(1) ANF
(2) Cortisol
(3) ADH
(4) ACTH

85) Choose the option with all incorrect events occuring during muscle contraction.

A.	Size of sarcomere	Decreases gradually and ultimately become less than size of the dark band
B.	Size of I band	Increases gradually
C.	Size of H zone	Remains same as that of dark band

(1) Only A and B
(2) Only B and C
(3) Only C
(4) A, B and C

86) The dorsal, flat, triangular body of scapula has a slightly elevated ridge called the spine which projects as a flat, expanded process called the

(1) Trochlear
(2) Acromion
(3) Glenoid
(4) Acetabulam

87) **Assertion :** Protoplasmic streaming is a type of movement exhibited by some organism.
Reason : Protoplasmic streaming can be observed in amoeba.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

88) The layer of connective tissue which surrounds each fasciculus is known as :-

(1) Sarcolemma
(2) Epimysium
(3) Perimysium
(4) Endomysium

89) Sacrum is present in which structure in human ?

(1) Pelvic girdle
(2) Vertebral column
(3) Pectoral girdle
(4) Fore limb

90) Parallely arranged filaments in the sarcoplasm of skeletal muscle fibre are called as :

(1) Muscle fibre
(2) Myofibrils
(3) Both 1 and 2
(4) Muscle tone

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	3	3	3	3	4	2	1	3	4	1	4	2	2	1	4	3	1	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	3	3	4	3	2	2	1	2	4	3	2	2	1	1	1	1	3	1	2
Q.	41	42	43	44	45															
A.	2	4	2	3	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	2	2	3	3	3	4	1	3	2	3	4	4	4	3	4	2	3	3	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	2	2	1	3	2	3	1	2	3	4	2	2	4	1	3	3	3	3	2
Q.	86	87	88	89	90															
A.	2	1	3	1	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	2	4	3	1	4	4	2	1	4	4	3	4	4	3	3	1	4	4	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	2	4	4	1	3	2	1	4	3	1	3	1	2	3	1	3	2	1	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	2	1	3	2	4	4	3	4	3	3	2	3	1	3	1	3	3	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	3	2	3	1	2	1	2	4	3	1	3	3	2	1	2	4	2	1	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	3	1	3	4	2	2	3	2	2										

SOLUTIONS

PHYSICS

1) Concept accelerates so the speed increases linearly with time. But the distance fallen increases as t^2. So the average speed occurs at half of time taken to pass the window, which is before it has covered half of the height of the window.

2) $\vec{v} = \frac{d\vec{r}}{dt}$

if $\vec{v} \cdot \vec{r}$ is +ive $|\vec{r}|$ increases with time $\vec{r} = x\hat{i} + y\hat{j}$

$\vec{v} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$

If $\vec{v} \cdot \vec{r}$ is –ve $|\vec{r}|$ decreases with time & particle is moving towards origin $\vec{v} \cdot \vec{r} < 0$

$x\frac{dx}{dt} + y\frac{dy}{dt} < 0$

3) $\frac{\sqrt{\left(\frac{R}{2}\right)^2 + H^2}}{T/2} = V_{av}$

$R = \frac{u^2 \sin(2\theta)}{g} = \frac{\left(4\sqrt{7}\right)^2 \sin(120^\circ)}{10}$

$= \frac{16 \times 7 \times \sqrt{3}/2}{10} = \frac{56\sqrt{3}}{10}m$

$H = \frac{u^2 \sin^2\theta}{2g} = \frac{\left(4\sqrt{7}\right)^2 \sin^2(60^\circ)}{2 \times 10}$

$= \frac{16 \times 7 \times \frac{3}{4}}{2 \times 10} = \frac{42}{10}$

$T = \frac{2u \sin\theta}{g} = \frac{2 \times 4\sqrt{7} \times \sin 60}{10}$

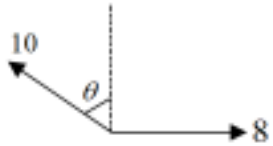
$= \frac{4\sqrt{7} \times \sqrt{3}}{10}$

$V_{average} = \frac{\sqrt{\left(\frac{28\sqrt{3}}{10}\right)^2 + \left(\frac{42}{10}\right)^2}}{\left(\frac{4\sqrt{7}\sqrt{3}}{10}/2\right)}$

= 7 m/s

4)

for shortest path



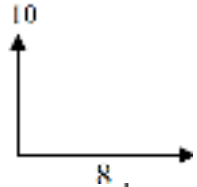
10 sinθ = 8

sinθ = 4/5

$t = \frac{d}{V_{MR}\cos\theta} = \frac{100}{10 \times 3/5}$

$= \frac{50}{3}\sec$

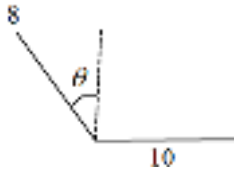
For minimum time



$t_{min} = \frac{d}{V_{MR}} = \frac{100}{10} = 10 \text{ sec}$

If $V_{MR} = 8$ and $V_R = 10$

then for zero drift

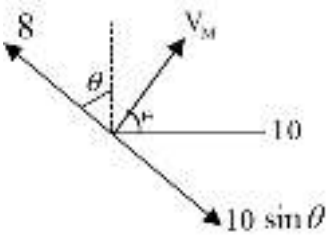


8 sinθ = 10

$\sin\theta = \frac{10}{8}$

Not possible

For minimum drift



10sinθ = 8

⇒ θ = 53°

Angle with down stream = 90 + 53

= 143°

5)

Minimum force on A = 12 N

So, both block to move together

$12 - \mu \times 4 \times 10 = 4 \times \frac{12}{9}$

$\Rightarrow \mu = \frac{1}{6}$

Now, maximum force F applied on lower block so, as to move together

$F - \mu 4 \times 10 = 5 \times \left(\frac{\mu \times 4 \times 10}{4}\right)$

⇒ F = μ × 10(5 + 4)

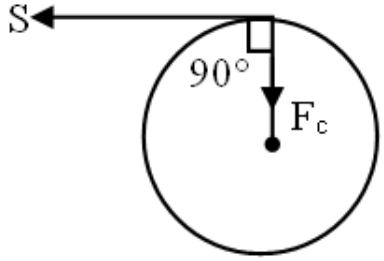
$= \frac{1}{6} \times 10 \times 9 = 15N$

6) $\nu = \sqrt{\mu gr} = \sqrt{0.5 \times 9.8 \times 40} = \sqrt{196} = 14 m/s$

7) Inertia means resistance to change. It is the property of the body by virtue of which it cannot change by itself its state of rest or of uniform motion.

8)

Mathematical Calculation
Since the angle between the force and displacement is 90°:

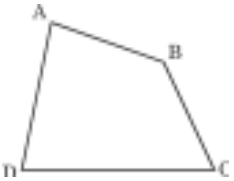


W = Fdcos90° = 0
Thus, for any motion along the circular path, the work done by the centripetal force is zero.
Final Answer
Correct Option: (2) Zero

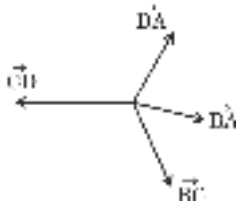
9) $U = -\int Fdx = -\int kxdx = -k\frac{x^2}{2}$
This is the equation of parabola symmetric to U axis in negative direction.

10)

Power $P_0 = F^aP^bM^c = M^{a+b+c}L^{a+b}T^{-2a-b} = M^1L^2T^{-3}$
$\Rightarrow a + b = 2$ ....(i)
$\Rightarrow a + b + c = 1$ ....(ii)
$\Rightarrow 2a + b = 3$ ....(iii)
$\Rightarrow c = -1 \Rightarrow a = 1 \Rightarrow b = 1$
$P_0 = FPM^{-1}$

11) 

$\vec{AB} + \vec{BC} + \vec{CD} + \vec{DA} = 0$
$\vec{BC} + \vec{CD} + \vec{DA} = -\vec{AB} = \vec{BA}$

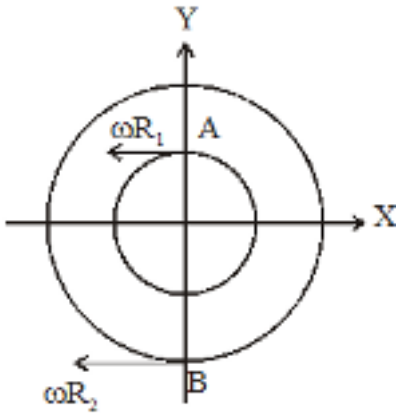


$$\vec{F}_R = \overrightarrow{BA} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DA} = \overrightarrow{BA} + \overrightarrow{BA}$$
$$\vec{F}_R = 2\overrightarrow{BA}$$

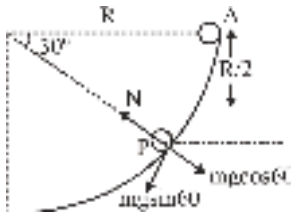
12) P = mv = 3.513 × 5.00 = 17.565 kg m/sec.
Ans. Should be rounding to 3 S.F.
⇒ P = 17.6 kg m/sec.

13)

$\theta = \omega t = \omega\frac{\pi}{2\omega} = \frac{\pi}{2}$



$$\vec{V}_A - \vec{V}_B = \omega R_1\left(-\hat{i}\right) - \omega R_2(-i)$$



14)
Apply work energy theorem at A & P
$$\frac{mv^2}{2} - 0 = mg \times \frac{R}{2}$$
Use dynamics equation at P, N –mg cos 60 = $\frac{mv^2}{R}$
Solving above equations, we get
⇒ N = mg + $\frac{mg}{2} = \frac{3mg}{2}$

15)

Suppose a be the linear acceleration of the mass and T the tension in the string.
Hence, Mg – T = Ma .....(i)
Let α be the angular acceleration of the flywheel. The couple applied to the flywheel is,

Iα = TR or T = $\frac{I\alpha}{R}$ .....(ii)

Now, we know that: a = Rα .....(iii)

Putting eqns. (ii) and (iii) in eqn. (i), we get;

$$Mg - \frac{I\alpha}{R} = MR\alpha$$

$$\alpha = \frac{MgR}{I + MR^2} \quad \text{.....(iv)}$$

$$= \frac{2 \times 9.8 \times 0.1}{0.1 + 2 \times (0.1)^2} = 16.66 \text{ rad/sec}^2.$$

16) $\vec{L}_{disc} = \vec{L}_{rotation} + \vec{L}_{translation}$

$= I\omega - mv\,(3R)$

$= \left(\frac{mR^2}{2}\right)\omega - m\,(R\omega)\,(3R)$

17)

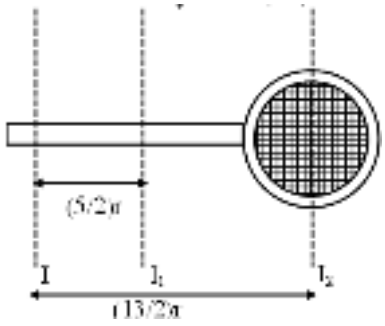
$I_1\omega_1 = I_2\omega_2$, $\omega_2 = \frac{1}{2}\omega_1$

$I_2 = 2I_1$

$$\frac{1}{2}m^0R^2 + (\mu t)R^2 = 2\left[\frac{1}{2}m_0R^2\right]$$

$$t = \frac{m_0}{2\mu}$$

18)



$$I = \left[I_1 + M\left(\frac{5}{2}r\right)^2\right] + \left[I_2 + M\left(\frac{13r}{2}\right)^2\right]$$

$$= \left[\frac{M\left(36r^2\right)}{12} + \frac{M\left(25r^2\right)}{4}\right] + \left[\frac{Mr^2}{2} + \frac{169Mr^2}{4}\right]$$

$= 52\ Mr^2$

19) $n_f = n_1 + n_2$

$$\left(P_0 + \frac{4T}{r}\right)r^3 = \left(P_0 + \frac{4T}{r_1}\right)r_1^3 + \left(P_0 + \frac{4T}{r_2}\right)r_2^3$$

$$\Rightarrow P_0\left(r^3 - r_1^3 - r_2^3\right) = 4T\left(r_1^2 + r_2^2 - r^2\right)$$

$$\frac{P_0}{\frac{4}{3}\pi}V = \frac{4T}{4\pi}(-S)$$

$\Rightarrow 3P_0V + 4ST = 0$

20) $v_1\rho_1 g = \frac{v_1}{2}\rho_w g$

$\rho_1 = \frac{\rho_w}{2}$ .....(i)

$v_2\rho_2 g = \frac{2}{3}v_2\rho_w g$

$\rho_2 = \frac{2}{3}\rho_w$ .... (ii)

(i)/(ii)

$\frac{\rho_1}{\rho_2} = \frac{3}{4}$

21) Pseudo force acts along direction opposite to acceleration of container.

$\Rightarrow P_{max}$ at point B

22) $v_1 = \sqrt{2g\frac{h}{2}} = \sqrt{gh}$

$\rho gh + 2\rho g\frac{h}{2} = \frac{1}{2}(2\rho)v_2^2$

$v_2 = \sqrt{2gh}$

$\frac{v_1}{v_2} = \frac{1}{\sqrt{2}}$

23)

Time to empty a vessel

$t \propto \left(\sqrt{\frac{2h_1}{g}} - \frac{2h_2}{g}\right)$

so $t \propto \sqrt{h}$ ....(1)

$t' \propto \sqrt{4h}$ ....(2)

$\frac{t}{t'} = \frac{1}{2} \Rightarrow \boxed{t' = 2t}$

24) **Solution:**

$V_{com} = \frac{m_1v_1 + m_2v_2}{m_1 + m_2}$

When both move opposition direction.

$V_{com1} = \frac{5 \times 1 - 2 \times 2}{1 + 2} = \frac{1}{3}$

when both move in same direction.

$V_{com2} = \frac{5 \times 1 + 2 \times 2}{1 + 2} = \frac{9}{3}$

$\frac{V_1}{V_2} = \frac{\frac{1}{3}}{\frac{9}{3}} = \frac{1}{9}$

25) $\lambda = a + \frac{bx}{L}$ (given)

$\lambda = \frac{dm}{dx} \Rightarrow dm = \lambda.dx = \left(a + \frac{bx}{L}\right) dx$

$X_{com} = \frac{\int_0^L x.dm}{\int_0^L dm} = \frac{\int_0^L x\left(a + \frac{bx}{L}\right) dx}{\int_0^L \left(a + \frac{bx}{L}\right).dx}$

$\frac{7L}{12} = \frac{\frac{aL^2}{2} + \frac{bL^2}{3}}{aL + \frac{bL^2}{2L}} \Rightarrow \boxed{2a = b}$

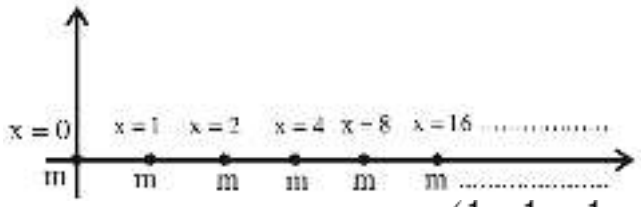
26) Let, F_1, F_2, F_4, F_8 ..... be the force of gravitation due to masses 'm' at x = 1, 2, 4, 8.... respectively.

$\Rightarrow F_1 = \frac{Gm^2}{1^2}$

$F_2 = \frac{Gm^2}{2^2}$

$F_4 = \frac{Gm^2}{4^2}$

$F_8 = \frac{Gm^2}{8^2}$



$F_1 + F_2 + F_4 + F_8.... = Gm^2 \underbrace{\left(\frac{1}{1} + \frac{1}{4} + \frac{1}{16} + \frac{1}{64}....\right)}_{\text{infinite G.P. with common ratio } = \frac{1}{4}}$

For an infinite G.P., sum $= \left(\frac{a}{1-r}\right)$

$\Rightarrow \text{Sum} = \frac{1}{1-\frac{1}{4}} = \left(\frac{4}{3}\right)$

a is the first term

r is the common ratio

$\Rightarrow F_1 + F_2 + F_4 + F_8..... = \frac{4}{3}Gm^2$

27)

$I = -\frac{dV}{dr}, \quad -K\int_{r_1}^{r_2} \frac{1}{r^2}dr = -\int_{V_1}^{V_2} dV$

$K\left[\frac{1}{r_1} - \frac{1}{r_2}\right] = V_2 - V_1$

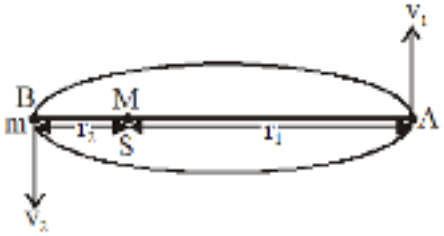
$6 \times 10^{-2}\left[\frac{1}{2 \times 10^{-2}} - \frac{1}{3 \times 10^{-2}}\right] = V_2 - 10$

$\therefore V_2 = 11$

28) According to conservation of angular momentum

$mv_1r_1 = mv_2r_2$

$$v_2 = \left(\frac{r_1}{r_2}\right) v_1$$



According to conservation of energy constant of energy

$E_A = E_B$

$$\frac{1}{2}mv_1^2 - \frac{GMm}{r_1} = \frac{1}{2}mv_2^2 - \frac{GMm}{r_2}$$

$$\frac{1}{2}m\left(v_1^2 - v_2^2\right) = \frac{1}{2}m\left[v_1^2 - \left(\frac{r_1}{r_2}\right)^2 v_1^2\right]$$

$$= GMm\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

Put value of v_2 and get

$$v_1 = \sqrt{\frac{2GMr_2}{r_1(r_1 + r_2)}}$$

then $J = mv_1r_1 = m\sqrt{\frac{2GM\,r_1r_2}{(r_1 + r_2)}}$

29)

$$\rho = \frac{m}{V} = \frac{\pi}{2 \times \frac{4}{3}\pi(0.5)^3}$$

$$= \frac{3}{8} \times \frac{1}{0.125} = 3\text{gm/cc}$$

Now $\eta = \frac{2}{9}\frac{r^2(\rho - \sigma)}{V_T}g \quad \left[V_T = \frac{S}{t}\right]$

$$= \frac{2}{9}\frac{(0.25)(3 - 1.2)(980)(5)}{50}$$

= 9.8 poise

30)

LC of vernier $= \left(\frac{10 - 9}{10}\right) \times 1.0\text{mm} = 0.10\text{mm}$

d = 0.4 mm, $Y = 2 \times 10^{11}\ \text{Nm}^{-2}$, L = 1 m

$$\because \Delta\ell = \frac{F\ell}{YA} = \frac{(1.5 \times 10)(1)}{\left(\pi \times 4 \times 10^{-8}\right)\left(2 \times 10^{11}\right)}$$

$\Delta\ell = 0.6 \times 10^{-3}$ m

$\Delta\ell$ = (L.C) (VSR)

0.6 mm = (0.1 mm) (VSR)
So, VSR = 6
So, 6th division of VS will coincide with main scale division.

31) $E_i = \frac{1}{2}mv_i^2 - \frac{GM_Em}{10R_E}$

$E_f = \frac{1}{2}mv_f^2 - \frac{GM_Em}{R_E}$

$\because E_i = E_f$

$\frac{1}{2}mv_i^2 - \frac{GM_Em}{10R_E} = \frac{1}{2}mv_f^2 - \frac{GM_Em}{R_E}$

$v_f^2 - \frac{2GM_E}{R_E} = v_i^2 - \frac{2GM_E}{10R_E}$

$v_f^2 = v_i^2 + \frac{2GM_E}{R_E}\left(1 - \frac{1}{10}\right)$

32)

Solution : For adiabatic process PV^{γ} = constant

Greater γ ⇒ steeper adiabatic curve.

$\gamma_{monoatomic} > \gamma_{diatomic} > \gamma_{polyatomic}$

Slope (m) : $m_3 > m_2 > m_1 \Rightarrow \gamma_3 > \gamma_2 > \gamma_1$
$\gamma_3 \rightarrow$ Monoatomic [Maximum]
$\gamma_2 \rightarrow$ Diatomic
$\gamma_1 \rightarrow$ Polyatomic

33) $Q = n C_p \Delta T$
$70 = 2 C_p(35–30)$
$C_p = 7$
$C_p – C_v = R$
$C_v = C_p – R = 7 – 2 = 5$
$C_v = 5$
$\theta = n C_v \Delta T$
$= 2 \times 5 \times 5 = 50$ cal

34)

$\Delta V = V\gamma_{app}\Delta T$

$\gamma_{app} = \frac{\Delta V}{V\Delta T}$

$\gamma_{app} = \frac{1}{100} \times \frac{1}{80}$

$\gamma_{app} = 1.25 \times 10^{-4}\ C^{-1}$

35)

Good absorber will behave as good emitter.
become $e_\lambda \propto a_\lambda$

36) $W_{net} = W_{AB} + W_{BC}$
$W_{AB} = (P \propto T) \quad (v = const.) \Rightarrow W_{AB} = 0$
$W_{BC} = 2 \times R \times 600 \times \square n \left(\frac{9}{3}\right)$
WBC = 1200 R(□n 3)

37) Given
x = A cos ωt
P.E. (U) = $\frac{1}{2} mw^2x^2 = \frac{1}{2} m\omega^2A^2\cos^2\omega t$
On plotting a graph between potential energy U and time t, we shall get a curve I and graph between U and x we shall get curve III.
P.E. is minimum at x = 0
P.E. is maximum at x = ±A

38) $\omega = \sqrt{\frac{k(m+M)}{mM}} = \frac{2\pi}{T}$
$T = 2\pi\sqrt{\frac{mM}{(m+M)k}}$

39) $T_□ = Ts$
$N2\pi\sqrt{\frac{\ell_\ell}{g}} = (N+1)\,2\pi\sqrt{\frac{\ell s}{g}}$
$N\sqrt{16\ell} = (N+1)\sqrt{\ell}$
N = 1/3
Time taken by longer pendulum
$T_\ell = N.2\pi\sqrt{\frac{\ell_\ell}{g}} = \frac{1}{3}2\pi\sqrt{\frac{16\ell}{g}}$
$= \frac{8\pi}{3}\sqrt{\frac{\ell}{g}}$

40) Here DL_1 = 40 m and
$DL_2 = \sqrt{40^2 + 9^2}$ = 41 m
Path difference Δx = 41 – 40 = 1 m
For first maximum $\Delta x = l = \frac{V}{f}$
$f = \frac{V}{\Delta x} = \frac{330}{1} = 330$ Hz

41) $v = \sqrt{\frac{Stress}{d}} = \sqrt{\frac{3 \times 10^8}{7.5 \times 10^5}} = \sqrt{400} = 20\,m/s$

42) The basic requirement for a wave function to represent a travelling wave is that for all values of x and t wave function must have a finite value. Out of the given function for y no one satisfied this condition. Therefore none can represent a travelling wave.

43) For first resonance :

Mean $L_2 = \frac{15.8 + 16.4}{2} = 16.1\text{cm}$

For second resonance :

Mean $L_2 = \frac{49.9 + 50.3}{2} = 50.1\text{cm}$

$\square_1$ = (16.1 – 0.2) cm = 15.9 cm

$\square_2$ = (50.1 – 0.2) cm = 49.9 cm

speed of sound = v = 2f ($\square_2$ – $\square_1$)

= $2(512)(49.9 - 15.9) \times 10^{-2}$

= 348.16 m/s ≈ 348 m/s

End correction :

$e = \frac{\ell_2 - 3\ell_1}{2} = \frac{49.9 - 47.7}{2} = 1.1\text{ cm}$

44)

$mc_1(T_1 - T_f) = mc_2(T_f - T_2)$

$c_1(T_1 - T_f) = c_2(T_f - T_2)$

Given : $c_1 = 0.8$ cal/g°C, $T_1 = 60°C$, $T_2 = 45°C$, $T_f = 53°C$

$0.8(60 - 53) = c_2(53 - 45)$

$0.8(7) = c_2(8)$

$5.6 = 8c_2$

$c_2 = \frac{5.6}{8} = 0.7\text{ cal/g°Cl}$

Final answer : Option (3)

45) $\frac{1}{z} = \frac{1}{x} + \frac{1}{y}$

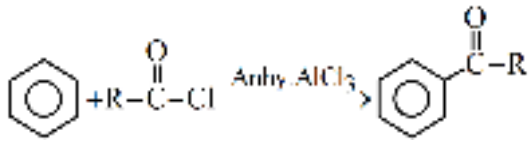
z = 66.7 cm

$\frac{1}{z^2}dz = \frac{1}{x^2}dx + \frac{1}{y^2}dy$

dz = 1.8 cm

z = (66.7 ± 1.8) cm

CHEMISTRY

46) 

47)

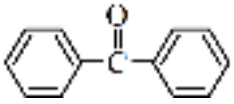
$Na_2S + Na_2[Fe(CN)_5NO]$
Sodium nitro prusside

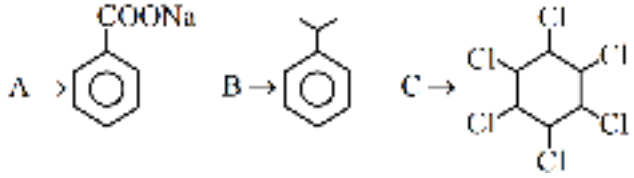
↓

$Na_4[Fe(CN)_5NOS]$
Violet colour
or
$[Fe(CN)_5NOS]^{4-}$

48) $$\%Br = \frac{80}{188} \times \frac{W_{AgBr}}{W_{org}} \times 100$$
$$= \frac{80}{188} \times \frac{0.94}{0.8} \times 100$$

49)

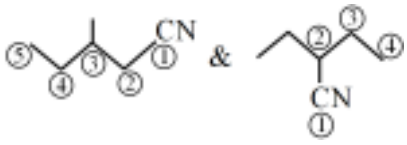
Product is 

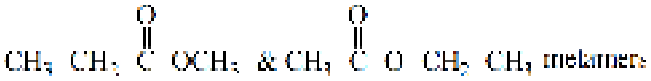
50) 

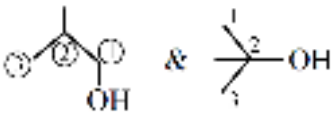
51)  nonplanar & non aromatic while other follows huckel's rule and are aromatic.

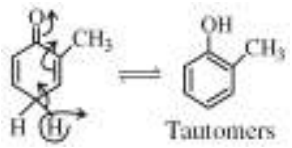
52)

Column chromatography &
Thin layer chromatography are example of adsorption chromatography where silica gel or Al_2O_3 acts as adsorbent while paper chromatography is an example of position chromatography where a special type of paper is used in which water molecules are trapped and act as stationary phase.

53)  Chain isomers

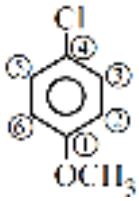
 metamers

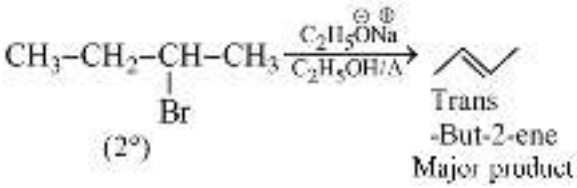
 position isomers



54)

Amides are very weak base because □.p. of nitrogen is effectively shared by oxygen.

55)  4-Chloro anisole

56) 

57)

Cis-isomer of But-2-ene has higher boiling point due to higher dipole moment.

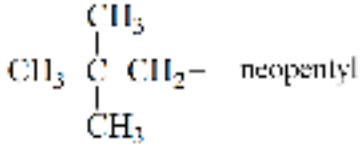
58)

$2H_2SO_4 \rightleftharpoons SO_3 + H_3O^+ + 4SO_4^{\ominus}$

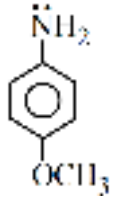
59)

(i) E.N. order sp-carbon > sp^2-carbon > sp^3-carbon

(ii)  THF

(iii)  neopentyl

All are correct

60) 

It is stronger base due to +m effect of $-OCH_3$

61)

Splitting of spectrum lines in presence of electric field is known as stark effect.

62)

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

Initial moles : 1 --

at eq^m : $1-\alpha$ 2α

Total moles = $1-\alpha + 2\alpha = 1 + \alpha$

Mole fraction : $\frac{1-\alpha}{1+\alpha}$ $\frac{2\alpha}{1+\alpha}$

$$\frac{(X_{NO_2}.P)^2}{(XN_2O_4)} = \frac{\left(\frac{2\alpha}{1+\alpha}\right).P^2}{\left(\frac{1-\alpha}{1+\alpha}\right).P} = \frac{4\alpha^2 P}{1-\alpha^2}$$

So, K_P =

$$\frac{K_P}{4P} = \frac{\alpha^2}{1-\alpha^2}$$

$K_P - K_P\alpha^2 = 4P\alpha^2$

$K_P = (K_P + 4P)\alpha^2$

$$\alpha = \sqrt{\frac{K_p}{4P + K_p}}$$

63)

$\rho^{d_{H_2O}} = 1$ g/mL

mass = 18 g

$$\text{moles} = \frac{18}{18} = 1$$

No. of neutrons = $1 \times N_A \times 8 = 8\ N_A$

64)

$\Delta G = \Delta H - T\Delta S$

for ΔH = +ve & ΔS = –ve

ΔG = +ve

Non-spontaneous

65)

pH = 12
$[H^+] = 10^{-12}$ mol/L
$[H^+]$ in 1mL = 10^{-15} mol
No. of H^+ ion = $10^{-15} \times 6.02 \times 10^{23}$
$= 6.02 \times 10^8$

66) The oxidation number of S are shown below along with the compounds

S_8	$S_2O_8^{-2}$	$S_2O_3^{-2}$	$S_4O_6^{-2}$
0	+6	+2	+2.5

Hence the order of increasing oxidation state of S is –
$S_8 < S_2O_3^{-2} < S_4O_6^{-2} < S_2O_8^{-2}$

67)

Molar mass of CH_4 = 12 + 4(1) = 16 g/mol.
The mass of N_A (Avogadro's number) molecules is 16 g.

Mass of one molecule = $\frac{\text{Molar mass}}{N_A} = \frac{16}{N_A}\text{g}.$

Mass percentage of carbon = $\left(\frac{\text{Mass of C}}{\text{Molar Mass of } CH_4}\right) \times 100\%$

Mass percentage of carbon = $\left(\frac{12}{16}\right) \times 100\% = 75\%$

68) $CO + H_2O \rightleftharpoons CO_2 + H_2$

3	3	x	0
2	2	(x + 1)	1

given $k = \frac{[CO_2][H_2]}{[CO][H_2O]}$

$5 = \frac{(x+1) \times 1}{2 \times 2}$

⇒ x + 1 = 20
⇒ x = 19

69) $\Delta S_r^o = \Sigma S_{Product}^o - \Sigma S_{Reactant}^o$
= 2 × 187 – 1 × 131 – 1 × 223
= 374 – 354
= 20 JK^{-1} mol^{-1}

70)

(A) In O_2 and O_3, all atoms are of the same element → oxidation number = 0 → **(A) is correct.**
(R) Oxygen has oxidation number −2 in most compounds, not in elemental forms or peroxide and superoxide → **(R) is not correct**

71) % hydrolysis of WAWB salt does not depend on concentration.

72)

$$T \propto \frac{n^3}{z^2}$$

$$\frac{T_1}{T_2} = \frac{n_1^3}{z_1^2} \times \frac{z_2^2}{n_2^3}$$

$$\frac{T_{He^+}}{T_{Li^{+2}}} = \frac{2^3}{2^2} \times \frac{3^2}{4^3} = \frac{2 \times 9}{64} = \frac{9}{32}$$

73) Equivalents of $C_2O_4^{2-}$ = Equivalents of MnO_4^-

x (mole) 2 = 1 5 ; $x = \frac{5}{2}$

74)

$$pH = \frac{1}{2}pK_a - \frac{1}{2}\log c$$

$$5.10 = \frac{1}{2}pK_a - \frac{1}{2}\log 10^{-2}$$

$$= \frac{1}{2}pK_a + 1$$

$pK_a = 2 \times 4.10 = 8.20$

K_a = Antilog (–8.20)

$= 6.4 \times 10^{-9}$

75)

	A	⇌	B	+	C
Initially	2		0		0
Moles at eq.	2 – x		x		x
	2 – 0.5		0.5		0.5

Since, two moles dissociated into 0.5

Therefore, one mole will dissociated into 0.25

76)

$CaCO_3 \rightarrow CaO(s) + CO_2(g)$

Molar mass of $CaCO_3$ = 40 (Ca) + 3 × 16 (O) = 100 g/mol

Molar mass of CaO = 40 (Ca) + 16 (O) = 56 g/mol

From 100 g $CaCO_3$, we get 56 g CaO residue.

So, from 200 g $CaCO_3$, CaO residue = $\frac{56}{100} \times 200$ = 112 g

77) St.-1 :- $E = -13.6 \times \frac{z^2}{n^2}$

For H-atom in n = 1 ; E = –13.6 eV

For He^+ in n = 2, $E = -13.6 \times \frac{(2)^2}{(2)^2}$
= –13.6 eV
St.1 is correct
St.-2 :- Incorrect (uncertainty principle)

78) Combustion reaction is exothermic and $\Delta G < 0$ for spontaneous process, $\Delta n_g > 0$ so $\Delta S > 0$.

79)

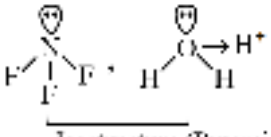
aA ⇌ □L + mM
On ↑ volume reaction moves towards more no. of moles.

80) $S = \sqrt{K_{sp}}$
For MnS, $S = \sqrt{10^{-16}} = 10^{-8}$ m It's maximum

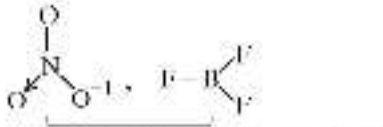
81)

Explain Question : Choose amphoteric combination of given elements.
Concept : This question is based on Zn, Al, Be, Ga, pb, Sn compounds are amphoteric.
Solution : A = Cr, B = Mn, C = Ga, D = O
□ $AD_3 = CrO_3$, $Ga_2O = C_2D$, BD = MnO Basic
acidic basic/Amphoteric
AD – CrO basic, $C_2D_3 = Ga_2O_3$ amphoteric
$A_2D_3 = Cr_2O_3$ amphoteric, $B_2D_7 = Mn_2O_7$ Acidic.
Final answer : 3

82)



Isostructure (Pyramidal)

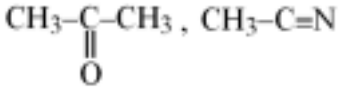


Isostructure (Trigonal planar)

83)

NH_3 – % s ch^r = 25%
PH_3 – % s ch^r = 100%
So $PH_3 > NH_3$ in % s ch^r of l.P.

84) NCERT, Pg. # 111 (Table - 4.6)

85) 

86) **Question Explanation:**
Question is asking about in which species bonds are non-directional

Concept:
Ionic Bond

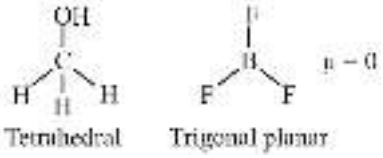
Answer Explanation:
RbCl is non-directional because it is an ionic molecule and all ionic compounds are non-directional.

Final Answer:
The correct option is (2).

87) K after loss of an electron attain inert gas configuration. Hence it will have maximum I.P.

88) $_{71}Lu$ – $[Xe]4f^{14}5d^{1}6s^{2}$
Period No = 6
Group No = IIIB
Block –f

89) **S-I :** hybrid orbitals are equal in energy and shape because they are formed from the same orbital.
S-II : hybrid orbital form stronger and more stable bond due to more directional nature.

90) 

BIOLOGY

91) NCERT Pg. # 5

92)

NCERT XI Page No. # 5

93) NCERT-XI, Page No. 6, 7, 8

94)

NCERT-XI_Pg. # 15

95)

NCERT-XI_Pg. # 16, 17, 18

96)

NCERT-XI_Pg. # 12,13,14,15

97)

NCERT-XI_ Pg. # 17, 18

98) *Chlorella*, *Chlamydomonas*, *Euglena*, *Gonyaulax*

99) NCERT-XI, Pg. No. # 11, 12, 13, 15, 18

100) NCERT Pg. # 27

101)

NCERT Pg. # 32, 33, 34

102) NCERT-XI, Pg. # 33

103) NCERT (XIth) Pg. # 32

104) NCERT XI, Pg # 25, 26, 27

105) NCERT-XI Pg. # 28, 31, 33

106) NCERT Pg. # 98, 100, 101

107) NCERT-XI Pg. # 102

108) NCERT-XI Pg. # 94

109) NCERT (XI) Pg # 95-96

110) NCERT Pg. # 101, 123, 124

111)

No. of chromosome in G_1 = 2n after S = 2n anaphase = 4n

112)

NCERT-XI_ Pg. # 123

113)

NCERT-XII, Pg. # 126-127

114)

NCERT Pg. # 128

115)

NCERT XI, Pg. No. # 123

116) NCERT Pg. # 106

117) NCERT XI, Pg. 108

118) NCERT Page No. 111, 112

119) NCERT, Pg. # 148, Modules

120) NCERT_Class-XII, Page # 109-110

121) NCERT pg. # 137

122) NCERT Pg. # 138, 139

123) The first bioassay of PGR was avena curvature test

124) Product and sites of the components of aerobic breakdown

125) NCERT XI Pg. # 154-160

126) NCERT (XII)th Pg. # 156, 159

127) NCERT Pg. # 162, 164

128) **Explain Question:** It asks how many cytochromes from the list are part of the respiratory electron transport system (ETS).
Concept : This question is based on ETS (Respiration)
Solution:
Cytochromes involved in mitochondrial respiratory ETS:
Cytochrome b
Cytochrome c_1
Cytochrome c
Cytochrome a
Cytochrome a_3
So, the correct answer is: 2. Four
Final answer: Option (2).

129) NCERT XI, Page # 233

130) NCERT XI Pg No. # 247

131)

NCERT_Pg. No. 176

132) NCERT Pg.#176

133) NCERT Pg.#177

134) NCERT Pg. # 167

135) NCERT, Pg. # 168

136)

NCERT XI Pg. No. # 51

137)

NCERT Pg. # 40

138) NCERT, Pg. # 49, 50

139) NCERT Pg. # 39

140)

NCERT XI Pg. No. # 49, 50, 51

141)

NCERT XI Pg. No. # 51

142) NCERT-XI, Pg # 101, Para.-7.1.1

143)

Module 4, Pg. No. # 88, 89

144)

Module- 4, Pg. No. # 125

145) NCERT XIth, Pg 112 (E), 112(H)

146)

Module 4, Pg. No. # 123

147) NCERT Pg. No.120/7.5.2

148) NCERT Page No. 82

149)

Module 4, Pg. No. # 160

150)

NCERT XI Pg. No. # 202

151)

NCERT XI Pg. No. # 200

152)

NCERT XI Pg. No. # 197

153)

NCERT XI Pg. No. # 196

154)

NCERT XI Pg. No. # 202

155)

NCERT XI Pg. No. # 189

156) NCERT, Pg # 186

157)

NCERT XI Pg. No. # 190

158)

NCERT XI Pg. No. # 187

159)

NCERT XI Pg. No. # 184

160) NCERT Pg.#207,208.

161) NCERT-XII, Pg. # 296

162)

NCERT XI Pg. No. # 207

163) NCERT XI Pg. # 210,211

164)

NCERT XI Pg. No. # 211

165)

NCERT XI Pg.No.321

166) NCERT, Pg. # 232-234

167)

NCERT XI Pg. No. # 214

168) NCERT Pg. # (E) 321, (H) 322

169) NCERT Pg.# 234,240,241

170)

NCERT XI Pg. No. # 243

171) NCERT Pg.# 335,336,337,339

172) NCERT Pg. # 248

173)

NCERT XI Pg. No. # 245

174)

NCERT XI Pg. No. # 247

175)

NCERT (XI) Pg. # 305

176)

NCERT XI Pg. No. #226

177)

NCERT XI Pg. No. # 217

178)

NCERT XI Pg. No. # 218

179)

NCERT XI Pg. No. # 225

180)

NCERT XI Pg. No. # 219